



**PARTICULATE MATTER CONTINUOUS EMISSION MONITORING SYSTEMS
RESPONSE CORRELATION AUDIT**

LA CYGNE GENERATING STATION - UNIT 1

SSID: 1070005

PREPARED FOR:

EVERGY

LA CYGNE GENERATING STATION

25116 EAST 2200TH ROAD

LA CYGNE, KANSAS 66040

PREPARED BY:

EVERGY TESTING & ANALYTICS

TOPEKA, KANSAS 66612

ET&A PROJECT 25-012

TEST DATES: 2/19-21/2025

REPORT DATE: 4/20/2025

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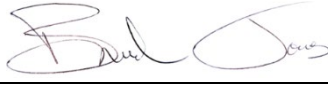
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TEST REPORT CERTIFICATION

This report, testing details, and approach have been developed under the supervision (including review) of the persons named below. Results contained in this report relate only to the sources tested and the parameters included in the test program.

Evergy Testing & Analytics (ET&A) operates as an air emission testing body (AETB) under a quality management system in conformance with ASTM D7036-04 "Standard Practice for Competence of Air Emission Testing Bodies".

Date: 4/20/2025

Signature 

Brandon Jones, QSTI
Manager of Environmental Testing
Evergy Testing & Analytics



Qualified Individual Statement

I, Brandon Jones, hereby agree on this day April 9, 2024, that as a qualified source testing individual, all projects conducted under my supervision will conform to the policies and procedures outlined in ASTM D 7036-04, 40CFR60, 40CFR75, all Federal, State, and local agencies, and the Evergy Testing & Analytics Quality Manual.


(Signature of Individual Signing Statement)

4/9/2024
(Date of Signature)

4/9/2024
(Date of QI Certification)

V
(Group of Certification)


(Signature of Supervisor/Witness)

4/9/2024
(Date of Signature)

INTRODUCTION

Facility Name and Location:

La Cygne Generating Station
25116 E. 2200th Rd
La Cygne, KS 66040

Reason for Conducting Test:

To conduct a Relative Correlation Audit (RCA) of the Particulate Matter Continuous Emissions Monitoring Systems (PM CEMS) at La Cygne Generating Station Unit 1 as required at 40CFR63, Subpart UUUUU.

Test Date(s):

February 19-21, 2025

Unit description:

Unit 1 is a coal fired boiler powering a steam turbine.

Testing Organization:

Evergy
Evergy Testing & Analytics
818 Kansas Ave
Topeka, KS 66612

Personnel Conducting Test:

Brandon Jones (QSTI: V) – Manager of Environmental Testing
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Evergy
Environmental Compliance Coordinator
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Phone: 913-402-4136

TEST RESULTS

RESULTS OF LINEAR CORRELATION

Plant/Unit: La Cygne Unit 1

Project No.: 25-012

Variable	Equation	Value
n	Number of data points =	14
$\bar{x}$	$\bar{x} = 1 / n * \sum x_i =$	6.07
S_{xx}	$S_{xx} = \sum (x_i - \bar{x})^2 =$	359
$\bar{y}$	$\bar{y} = 1/n * \sum y_i =$	5.79
S_{yy}	$S_{yy} = \sum (y_i - \bar{y})^2 =$	348
S_{xy}	$S_{xy} = \sum ((x_i - \bar{x}) * (y_i - \bar{y})) =$	351.3
b_0	$b_0 = \bar{y} - b_1 \bar{x} =$	-0.1551
b_1	$b_1 = S_{xy} / S_{xx} =$	0.9793
S_L	$S_L = \sqrt{1 / ((n-2) * \sum (y_i^{\wedge} - y_i)^2)} =$	0.592
$\hat{y}_{\text{mean}}$	$\hat{y}$ at mean x value =	5.786
t_f	t_f from table =	2.179
CI	$CI = t_f * S_L * \sqrt{1 / n} =$	0.345
EL	Emission Limit =	10.18
CI%	$CI\% = (CI / EL) * 100 =$	3.38%
n'	$n' = n =$	14
v_f	v_f from table =	1.515
u_n	u_n from table =	1.186
k_T	$k_T = u_n * v_f =$	1.798
TI	$TI = k_T * S_L =$	1.064
TI%	$TI\% = (TI / EL) * 100 =$	10.4%
S_y	$S_y = \sqrt{(S_{yy} / (n - 1))} =$	5.176
r^2	$r^2 = 1 - (S_L^2 / S_y^2) =$	0.987
r	$r = \sqrt{1 - (S_L^2 / S_y^2)} =$	0.993

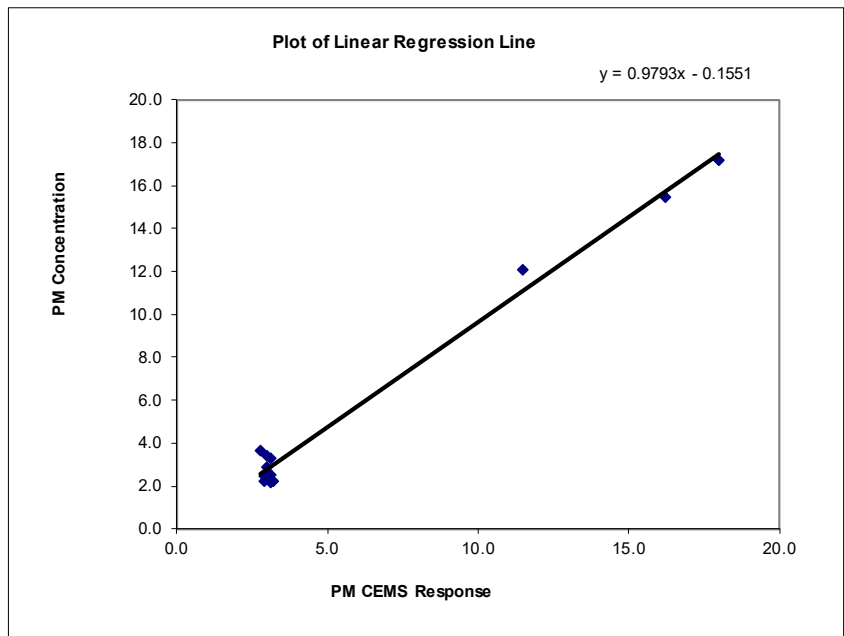
* Indicates correlation coefficient is undefined.

Correlation equation: $y = 0.9793 x - 0.1551$

Summary of Acceptance Criteria for PS-11

Criterion	Actual	Allowable	Acceptable?
Correlation coefficient	0.993	≥ 0.85	yes
Confidence interval	3.38%	$\leq 10\%$	yes
Tolerance interval	10.4%	$\leq 25\%$	yes

* Indicates correlation coefficient is undefined.



TESTING METHODOLOGY

The CEMS referenced in this report met all performance criteria for certification. The performance specification includes the periods during which the CEMS relative accuracy testing was conducted.

The following section describes the procedures used for the RCA. The test was conducted in such a way that it yielded measurements representative of the emissions from the source and could be correlated to the plant CEMS data. The RCA testing consisted of sampling for concentrations of particulate matter in the flue gas at the Unit 1 exhaust stack.

During the RCA 14 runs were performed. Up to two (2) test runs could be discarded, as is allowed by 40 CFR 60, Appendix F, Procedure 2, for a total of 12 runs. Two (2) test runs were discarded.

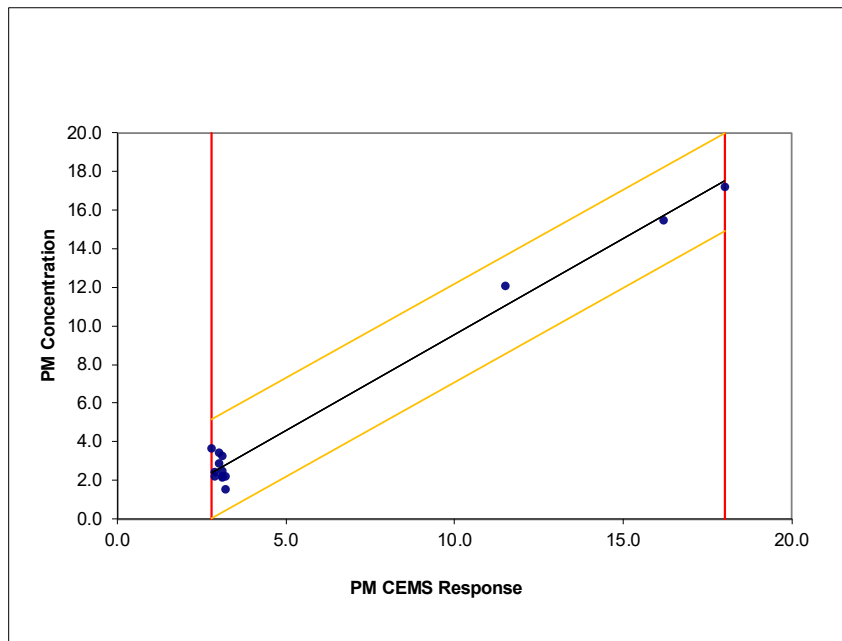
The following presents a graphical representation of the test runs plotted against the PM CEMS measurements collected during the test periods. A linear regression was plotted from the data set. The correlation coefficient was greater than 0.85, the confidence interval was less than 10%, and the tolerance interval was less than 25%. All PS11 criteria satisfied.

CORRELATION TEST PM CEMS AND REFERENCE METHOD TEST DATA

Mark X for Omit	Run	PM CEMS	PM concentration
		x	y
	1	2.900	2.437
	2	3.000	2.874
	3	2.800	3.638
	4	3.100	3.253
	5	3.100	2.490
	6	3.000	3.412
	7	2.900	2.230
	8	3.100	2.175
	9	3.200	2.225
	10	11.500	12.064
	11	16.200	15.458
	12	18.000	17.176
x	13	3.100	2.208
x	14	3.200	1.543

Plant/Unit: La Cygne Unit 1
Project No.: 25-012

Emission limit (mg/acm): 10.18
Maximum PM CEMS (mg/acm): 18.000
Minimum PM CEMS (mg/acm): 2.800



REFERENCE METHOD AND RELATIVE ACCURACY PROCEDURES

Every Testing & Analytics personnel tested in accordance with the following U.S. EPA source emissions test methods referenced in *40 CFR 60, Appendix A and 40 CFR 63, Subpart UUUUU*.

- Method 1 – Sample and Velocity Traverses for Stationary Sources.
- Method 2 – Determination of Velocity and Volumetric Flow Rate in Stationary Sources
- Method 3A – Determination of Oxygen and Carbon Dioxide Concentrations in Emissions from Stationary Sources (Instrumental Analyzer Procedure)
- Method 4 – Determination of Moisture in Emissions from Stationary Sources
- Method 5 (Modified for MATS) – Determination of Particulate Matter Emissions from Stationary Sources

SAMPLING AND ANALYTICAL PROCEDURES

USEPA Method 1 was used to determine the appropriateness of the existing ports as the sampling location and to determine the number of sampling points to be used when sampling at this location.

Upstream Duct Diameters	Downstream Duct Diameters	Number of Ports	Number of Point per Port	Distance from Far Wall to outside of Port	Port length	Depth of Duct
4.7	8.7	4	3	383.0	8	375.0

After determining the number of sampling points, a Type “S” pitot tube such as the one defined in USEPA Method 2 was used to determine if cyclonic flow conditions existed at the sampling location. ET&A leveled and zeroed a manometer and connected a Type “S” pitot tube and leak-checked the system. The pitot tube was positioned at each traverse point, in succession, so that the planes of the face openings of the pitot tube were perpendicular to the stack cross-sectional plane. This position was referred to as “0° reference point.” The differential pressure (ΔP) reading at each traverse point was recorded. The pitot tube was rotated until a null reading was obtained. The value of the rotation angle was determined to the nearest degree using an angle finder. After this technique was applied at each traverse point, the average of the absolute value of the angles was determined. The average angles were ≤ 20 degrees and the overall flow conditions were found to be acceptable.

After determining the number of sampling points to use, USEPA Method 2 was used to perform preliminary velocity measurements of the flue gas in the stack while checking the system for flow conditions. Preliminary velocity measurements were made in each stack to determine the appropriate nozzle size to use to maintain isokinetic sampling rates during the sampling. Velocity measurements were also made while collecting the samples to ensure the samples were collected isokinetically and to determine the pollutant concentrations and mass emission rates.

The stack gas velocity was determined from the measurement of the velocity head at each traverse point, with a calibrated Type “S” pitot tube connected to an inclined manometer. Measurement of the stack gas temperature was made with a calibrated K Type thermocouple, and determination of the stack gas composition was made using USEPA Method 3A.

ET&A made measurements of the face opening alignments, external tubing diameter, and base-to-opening plane distances of the pitot tubes. These measurements meet the design criteria in USEPA Method 2 for a Type “S” pitot tube and, therefore, a baseline coefficient value of 0.84 was assigned to the pitot tube. Verification of these measurements was recorded onto a pitot tube calibration worksheet and presented as part of the final report. Once the measurements of the pitot tube were completed, the system was assembled and leak-checked prior to making any measurements.

In addition to the velocity head measurements, USEPA Method 3A was used to determine O_2 and CO_2 concentrations in the flue gas. The O_2 and CO_2 concentrations were made to calculate the flue gas molecular weight. The latter measurements were also made to complete the preliminary velocity calculations necessary to establish the sampling conditions for setting up the actual pollutant measurement test runs.

The Method 3A sample was extracted from the stack through a heated stainless-steel probe into a chilled moisture knockout system using ¼ -inch PTFE sample line and a regulated K&F single diaphragm sample pump. The sample was then stored in a Tedlar Bag for analysis. The bag sample was then analyzed for a set amount of time in order to record a stable reading. The analyzer output was recorded continuously by a RATAView Data Acquisition System (DAS) manufactured by ESCSpectrum.

USEPA Protocol-1 calibration gases were used to calibrate the analyzer system. A calibration error test was performed on each analyzer by introducing a zero, mid and high-level calibration gas directly to the analyzer inlet. A zero gas (nitrogen) of less than 0.25 percent of span, a certified Protocol-1 mid-level (40 to 60 percent) of span and a certified Protocol-1 high-level (100 percent) of span were used for the analyzer calibration error test.

The measurement system was required to meet the following QA/QC criteria for this test program:

- Analyzer Calibration Error: ± 2.0 percent of the calibration span for the zero, mid and high-level calibration gases.
- System Bias: ± 5.0 percent of the analyzer calibration span for the zero and upscale calibration gases.
- Calibration Drift: ± 3.0 percent of the analyzer calibration span for the zero and mid or high-level gases.

REFERENCE METHOD CEMS CALIBRATION GASSES

Pollutant	Level	Cylinder #	Certified Value
CO ₂ / O ₂	Zero	CC246055	Zero
	Mid	CC199274	9.61 / 9.98
	High	DT0031874	20.10 / 20.00

ANALYZER INFORMATION AND TECHNOLOGY

Brand	Model	Compounds Measured	Measurement Technology	Range	Sample Condition
Teledyne	T802	CO ₂ /O ₂	NDIR / Paramagnetic	0-25 %	Cold, Dry

For this testing, the Method 4 sampling train was incorporated into the Method 5 sampling trains. The moisture content of the sample gas stream was determined by extracting the gas sample at a known and regulated rate through the glass condenser train maintained below 68 degrees Fahrenheit (°F) in an ice bath with a vacuum pump. The volume of gas sampled was measured with a calibrated dry gas meter. The volume of moisture collected was measured gravimetrically with an analytical balance with a tolerance of 0.1 gram (g).

METHOD 5

The sampling and analytical procedures used during the emission testing for PM were conducted in accordance with the procedures outlined in USEPA Method 5. The probe and filter exit temperature was maintained at 320 ± 25 °F, in accordance with the requirements of the test method. The sampling equipment consisted of the following equipment and components:

1) Probe assembly

- a) Nozzle – Calibrated borosilicate glass with a sharp, tapered leading edge.
- b) Probe - Stainless steel sheath with a 5/8-inch diameter glass insert wrapped with nichrome wire; electronically controlled and capable of maintaining 320 ± 25 °F with a calibrated K Type thermocouple.
- c) Pitot - Calibrated stainless-steel Type "S" pitot tube constructed and attached to the probe according to the specifications outlined in USEPA Method 2, 40 CFR 60.
- d) Thermocouple - Calibrated K Type thermocouple attached to the sample probe such that the tip has no contact with any metal and does not interfere with the pitot tube face openings and the sample nozzle.

2) Filter Box Assembly

- a) Filter Holder - Borosilicate glass, with a Teflon frit filter support and a silicone rubber gasket.
- b) Filter Heating System – The heating system is capable of monitoring and maintaining the temperature around the filter to ensure the sample gas temperature exiting the filter is 320 ± 25 °F during sampling.
- c) Filter Temperature Sensor – A thermocouple capable of measuring temperature to within ± 3 °C (5.4 °F) is installed so that the sensing tip of the temperature sensor the temperature of the filter. The filter temperature sensor was monitored and recorded during sampling to ensure a sample gas temperature exiting the filter of 320 ± 25 °F.

3) Impingers

- a) The back-half Method 5 impingers consisted of four sequential impingers. Impingers one and two contained 100 ml each of DI water. Impinger three was empty to prevent liquid carry over. The final impinger contained at least 300 grams of silica gel.
- b) The first and third impingers were of the Modified Greenburg-Smith design. The second impinger was of the Greenburg-Smith design with a standard orifice tip. Final gas exit temperature was measured with a calibrated K Type thermocouple immersed in the gas stream.

4) Control Box

- a) The control box is a module containing a vacuum gauge, external leak-free pump, thermocouples capable of measuring temperature within ± 5 °F, dry gas meter with a minimum of 2 percent accuracy, valves and related equipment to determine sample gas volume.

The control box was leak-checked from pump to orifice at a minimum of 15 inches of mercury (in. of Hg) vacuum. The selected nozzle was attached to the sample probe. The moisture sample train was prepared per Method 4. The cold box portion was packed with crushed ice to maintain final gas exit temperature below 68 °F.

A pre-weighed quartz fiber filter, prepared per Method 5, was placed inside the filter holder onto a Teflon® support frit. The filter holder was positioned into the filter assembly unit and connected to the sample probe and impinger train with interconnecting glassware. The entire assembled sample system was leak-checked by plugging the inlet to the probe nozzle and pulling a vacuum of at least 15 inches of Hg. A leak rate less than 0.02 cubic feet per minute (cfm) was determined and considered acceptable as outlined in 40 CFR 60. The pitot tube system was also vacuum and pressure leak-checked at greater than 3 inches of water column. The traverse sampling points were marked on the probe for easy visibility.

Temperature controllers for the probe were set to maintain a temperature of $320 \pm 25^{\circ}\text{F}$. The nozzle was placed at the first traverse point and positioned directly facing the stack gas stream. The sampling pump was started and the meter set to the appropriate sampling rate at each of the individual traverse points. At the conclusion of each test run, the sample pump was turned off, the probe was removed from the stack, and the final sample train measurements were recorded. A final leak-check of the system was performed at the conclusion of every test run, as previously described at the highest vacuum encountered during testing. A leak-check of the pitot system was performed as previously described. The final leak-checks were determined to be within the guidelines outlined in Method 5. All velocity measurements, sample train measurements, moisture data, and leak-check measurement were recorded onto the field data worksheets.

The PM filter holder and the interconnecting hardware were removed from the sample probe and the openings sealed. Likewise, the sample probe was disconnected. The impinger train, sample probe, PM filter holder, and interconnecting glassware were relocated to the sample collection area. The PM filter holder was labeled with sample identification and prepared for transfer to the ET&A's on-site laboratory. The nozzle was carefully removed from the sample probe and rinsed with acetone into a glass container. Acetone was used to rinse the probe liner, ferrules, nozzle, and all interconnecting glassware utilized between the sample nozzle and filter holder. A nylon brush was used to loosen any adhering PM and subsequent washes and rinses were collected into the sample container. Acetone was also used to rinse any PM adhering to the nylon brush into the sample collection container. An acetone blank sample was collected at the sample cleanup area at the facility. The blank sample was analyzed using the same procedures as the test samples. The acetone blank was found to be within specifications and did not impact the results.

Prior to conducting the final analysis, a preliminary analysis was conducted in the field following sample recovery in order to give real time performance indicators to facility personnel. Upon arrival at the ET&A on-site mobile laboratory, the integrity of the PM filter holders was inspected and found to be intact. The quartz fiber filter was removed from the filter holder and placed into the original petri dish. Any excess PM contained within the front half of the filter holder was brushed into the petri dish. The front half of the filter holder was rinsed and cleaned with the nylon brush and the washes were collected into the appropriate sample container with the other sample train washes for each respective test run.

The preliminary particulate analysis was conducted by ET&A in accordance with USEPA Method 5 with shortened desiccation periods and intervals between weights. The filters were transferred into the desiccator for an hour to obtain a weight and then weighed every 30 minutes, minimum, until two consecutive weights were obtained that agreed within ± 0.5 milligrams.

The acetone rinses and an acetone blank were evaporated in tared beakers on a hot plate to determine the level of particulate collected in the nozzle, probe liner and the front half of the filter holder and to determine the amount of residual weight each beaker retained due to acetone impurities. After evaporating to dryness, the beakers were transferred to a desiccator for one hour to obtain a weight and then weighed every 30 minutes, minimum, until two consecutive weights were obtained that agreed within ± 0.5 milligrams. Each filter, acetone washing, and acetone blank was individually weighed on an analytical balance with a sensitivity of 0.1 milligram.

The final particulate analysis was conducted by ET&A of Topeka, Kansas in accordance with USEPA Method 5. The filters and the beakers containing the particulate recovered from the probe and nozzle rinses were transferred into the desiccator for twenty-four hours to obtain a weight and then weighed every six hours, minimum, until two consecutive weights were obtained that agreed within ± 0.5 milligrams.

The flue gas moisture collected in the first three impingers was measured and recorded. The moisture laden silica gel in the fourth impinger was weighed. The weight gain of the silica gel was added to the measured moisture condensed during the test run to determine the total moisture collected for that run.

QUALITY ASSURANCE / QUALITY CONTROL SUMMARY

ET&A has established quality assurance and quality control (QA/QC) guidelines for providing quality sampling and analytical data from source tests. These QA/QC procedures were implemented to ensure the acceptability and reliability of the data generated. In summary, an appropriate degree of data quality was maintained throughout this project. The sampling trains were leak checked prior to and immediately after sampling. Leak rates for the isokinetic sampling trains were less than the maximum criterion of 0.02 cubic feet per minute.

The isokinetic sampling rates for the isokinetic sampling trains were within the $100\% \pm 10\%$ criterion established for isokinetic sampling. The pitot leak checks were stable for at least 15 seconds. USEPA Method 5 requires the calibration of the metering system after each field use. This is typically performed in a laboratory environment. If it fails the test, the test results are invalidated. This makes a field calibration procedure highly desirable. USEPA's approved alternative method (ALT-009) based on the pre-test meter coefficient found in Method 5 Section 4.4.1 has been approved as a post-test calibration procedure. The procedure states that, "If the average of the three runs is within $\pm 5\%$ of the y-value, the Method 5 sampling system passes the post calibration requirements."

DATA UNCERTAINTY

Both qualitative and quantitative factors contribute to field measurement uncertainty and should be taken into consideration when interpreting the results presented in this test report. There are several factors that can affect qualitative and quantitative measurements.

Qualitative uncertainty factors include, but are not limited to, unknown chemical interferences, sample matrix interactions, environmental conditions, sample handling and instrument operation and maintenance. To reduce the impact of these qualitative uncertainty factors, ET&A has developed a set of Standard Operating Procedures (SOPs) in accordance with our corporate quality assurance guidelines and ASTM D 7036-04.

Quantitative uncertainty factors known to directly affect uncertainty include the accuracy of calibration standards as well as the precision and accuracy of instrument measurements and the test methods utilized. To reduce the impact of these quantitative uncertainty factors, ET&A utilizes testing and analytical methodology that has been approved by EPA or the American Society for Testing and Materials (ASTM) where applicable. In addition, ET&A personnel perform routine instrument and equipment calibrations according to manufacturer's guidelines and/or test method specifications.

The limitations of the various methods, instruments, equipment, and materials utilized during this project have been reasonably considered to be in accordance with the project data quality objective, but the ultimate impact of the cumulative uncertainty of this project is not fully identified within the results of this test report.

RELATIVE CORRELATION AUDIT

RESULTS OF LINEAR CORRELATION

Plant/Unit: La Cygne Unit 1

Project No.: 25-012

Variable	Equation	Value
n	Number of data points =	14
$\bar{x}$	$\bar{x} = 1 / n * \sum x_i =$	6.07
S_{xx}	$S_{xx} = \sum (x_i - \bar{x})^2 =$	359
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b_0	$b_0 = \bar{y} - b_1 \bar{x} =$	-0.1551
b_1	$b_1 = S_{xy} / S_{xx} =$	0.9793
S_L	$S_L = \sqrt{1 / ((n-2) * \sum (y_i - \hat{y}_i)^2)} =$	0.592
$\hat{y}_{\text{mean}}$	$\hat{y}$ at mean x value =	5.786
t_f	t_f from table =	2.179
CI	$CI = t_f * S_L * \sqrt{1 / n} =$	0.345
EL	Emission Limit =	10.18
CI%	$CI\% = (CI / EL) * 100 =$	3.38%
n'	$n' = n =$	14
v_f	v_f from table =	1.515
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S_y	$S_y = \sqrt{S_{yy} / (n - 1)} =$	5.176
r^2	$r^2 = 1 - (S_L^2 / S_y^2) =$	0.987
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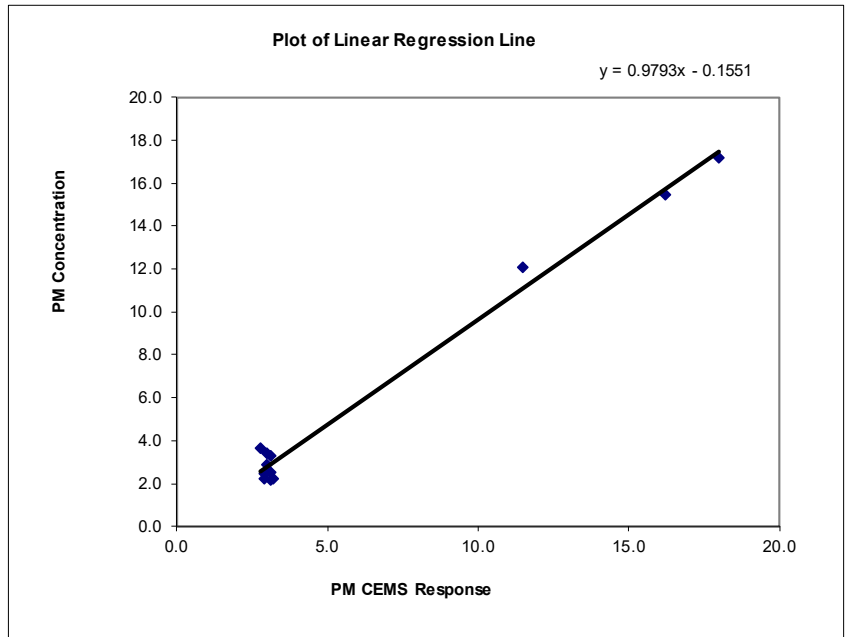
* Indicates correlation coefficient is undefined.

Correlation equation: $y = 0.9793 x - 0.1551$

Summary of Acceptance Criteria for PS-11

Criterion	Actual	Allowable	Acceptable?
Correlation coefficient	0.993	≥ 0.85	yes
Confidence interval	3.38%	$\leq 10\%$	yes
Tolerance interval	10.4%	$\leq 25\%$	yes

* Indicates correlation coefficient is undefined.



CALCULATIONS FOR LINEAR CORRELATION

Plant/Unit: La Cygne Unit 1
Project No.: 25-012

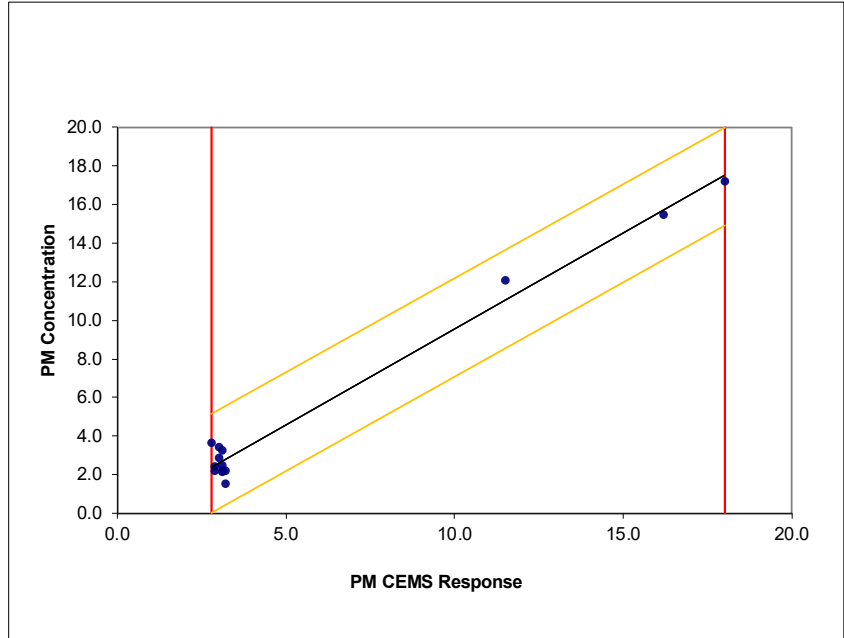
Run	PM CEMS response	PM concentration	CALCULATED STATISTICAL PARAMETERS				
	x	y	$(x_i - \bar{x})^2$	$(y_i - \bar{y})^2$	$(x_i - \bar{x}) * (y_i - \bar{y})$	$y^{\wedge}$	$(y^{\wedge}_i - y_i)^2$
1	2.9	2.4	10.0278	11.2141	10.6044	2.6849	0.0613
2	3.0	2.9	9.4044	8.4799	8.9302	2.7828	0.0083
3	2.8	3.6	10.6711	4.6152	7.0178	2.5870	1.1041
4	3.1	3.3	8.8011	6.4178	7.5156	2.8808	0.1383
5	3.1	2.5	8.8011	10.8618	9.7773	2.8808	0.1524
6	3.0	3.4	9.4044	5.6342	7.2792	2.7828	0.3963
7	2.9	2.2	10.0278	12.6437	11.2600	2.6849	0.2067
8	3.1	2.2	8.8011	13.0364	10.7114	2.8808	0.4975
9	3.2	2.2	8.2178	12.6841	10.2095	2.9787	0.5687
10	11.5	12.1	29.5211	39.4114	34.1097	11.1069	0.9157
11	16.2	15.5	102.6844	93.5497	98.0107	15.7097	0.0633
12	18.0	17.2	142.4044	129.7252	135.9171	17.4724	0.0880
Sum	72.80	69.43	358.76667	348.27359	351.34293		4.20078
Mean	6.07	5.79		$y^{\wedge}$ at mean x =		5.79	

CORRELATION TEST PM CEMS AND REFERENCE METHOD TEST DATA

Mark X for Omit	Run	PM CEMS	PM concentration
		x	y
	1	2.900	2.437
	2	3.000	2.874
	3	2.800	3.638
	4	3.100	3.253
	5	3.100	2.490
	6	3.000	3.412
	7	2.900	2.230
	8	3.100	2.175
	9	3.200	2.225
	10	11.500	12.064
	11	16.200	15.458
	12	18.000	17.176
x	13	3.100	2.208
x	14	3.200	1.543

Plant/Unit: La Cygne Unit 1
Project No.: 25-012

Emission limit (mg/acm): 10.18
Maximum PM CEMS (mg/acm): 18.000
Minumim PM CEMS (mg/acm): 2.800





EPA Method 5 : Determination of Particulate Emissions from Stationary Sources

Field Data

	Run #1	Run #2	Run #3	Run #4	Run #5
Date:	2/19/2025	2/19/2025	2/19/2025	2/19/2025	2/19/2025
Start time:	10:34	12:00	13:12	14:48	16:00
Stop time:	11:37	13:03	14:15	15:51	17:03
Sample volume (ft ³):	46.748	37.776	36.805	37.428	37.190
Sampling time (minutes):	48	48	48	48	48
Stack temp. (°F):	129	129	129	129	129
Meter temp. (°F):	59	63	66	68	70
Barometric pressure (in Hg):	29.23	29.21	29.19	29.16	29.15
Static pressure (in H ₂ O):	-0.75	-0.89	-0.95	-0.98	-0.98
Stack pressure (in H ₂ O):	29.17	29.14	29.12	29.09	29.08
Oxygen (%dry):	7.6	7.1	7.0	7.0	7.1
Carbon Dioxide (%dry):	11.9	12.5	12.6	12.5	12.5
Moisture (gm):	164.2	133.3	140.9	140.7	142.5
ΔH (in H ₂ O):	3.283	2.150	2.050	2.083	2.075
√Δp (in H ₂ O):	0.7994	1.0344	1.0059	1.0095	1.0023
DGM Y _c :	0.986	0.986	0.986	0.986	0.986
Pitot Tube Coefficient (C _p):	0.84	0.84	0.84	0.84	0.84
Nozzle diameter (inches):	0.286	0.226	0.226	0.226	0.226
Stack diameter (inches):	375.00	375.00	375.00	375.00	375.00
F _c :	1840	1840	1840	1840	1840

Method 5 Laboratory Data

Beaker Tare Weight (g):	51.7641	51.7811	53.2737	51.9409	53.5973
Beaker Final Weight (g):	51.7688	51.7858	53.2799	51.9464	53.6012
Filterable PM mass (probe) (g):	0.0047	0.0047	0.0062	0.0055	0.0039
Filter Tare Weight (g):	0.3528	0.3522	0.3547	0.3536	0.3579
Filter Final Weight (g):	0.3536	0.3527	0.3549	0.3539	0.3584
Filterable PM mass (filter) (g):	0.0008	0.0005	0.0002	0.0003	0.0005
Acetone Wash (ml):	88	62	60	59	65

Calculations

Maximum allowed blank correction (g):	0.00069	0.00049	0.00047	0.00046	0.00051
Acetone blank PM, if allowed (g):	0.00000	0.00000	0.00000	0.00000	0.00000
Blank corrected PM (g):	0.00550	0.00520	0.00640	0.00580	0.00440
Sample volume (dscf) (Vmstd):	46.204	36.939	35.717	36.155	35.811
Moisture volume (scf) (Vwstd):	7.742	6.285	6.643	6.634	6.719
Moisture Fraction Measured (BWSmsd):	0.144	0.145	0.157	0.155	0.158
Moisture Fraction @ Saturation (BWSsat):	0.149	0.150	0.150	0.150	0.150
Moisture Fraction (BWS):	0.144	0.145	0.150	0.150	0.150
Molecular weight (dry) (Md):	30.21	30.29	30.29	30.29	30.28
Molecular weight (actual/wet) (Ms):	28.46	28.50	28.36	28.38	28.34
Gas velocity (ft/sec) (Vs):	48.3	62.5	61.0	61.2	60.8
Stack area ft ² (As):	766.99	766.99	766.99	766.99	766.99
Gas flow (acfm) (Qa):	2,224,248	2,877,401	2,805,996	2,816,822	2,799,036
Gas flow (dscfm) (Qs):	1,666,627	2,149,031	2,083,605	2,088,923	2,074,894
% Isokinetic:	99.3	98.6	99.1	99.9	99.9
Filterable PM (mg/acm):	2.4373	2.8740	3.6377	3.2527	2.4903



EPA Method 5 : Determination of Particulate Emissions from Stationary Sources

Field Data

	Run #6	Run #7	Run #8	Run #9	Run #10
Date:	2/19/2025	2/20/2025	2/20/2025	2/20/2025	2/21/2025
Start time:	17:07	13:00	15:26	17:51	10:55
Stop time:	18:10	15:15	17:41	20:06	11:53
Sample volume (ft ³):	37.688	92.584	90.519	94.443	35.858
Sampling time (minutes):	48	120	120	120	48
Stack temp. (°F):	129	129	129	129	124
Meter temp. (°F):	70	70	74	75	64
Barometric pressure (in Hg):	29.15	29.24	29.19	29.19	29.25
Static pressure (in H ₂ O):	-0.99	-0.85	-0.95	-0.90	-0.72
Stack pressure (in H ₂ O):	29.08	29.18	29.12	29.12	29.20
Oxygen (%dry):	7.1	6.8	7.0	7.1	9.3
Carbon Dioxide (%dry):	12.4	12.8	12.6	12.4	10.4
Moisture (gm):	143.1	339.1	339.2	350.2	126.1
ΔH (in H ₂ O):	2.117	2.079	1.963	2.133	1.908
√Δp (in H ₂ O):	1.0091	1.0027	1.0145	1.0111	0.7488
DGM Y _c :	0.986	0.986	0.986	0.986	0.986
Pitot Tube Coefficient (C _p):	0.84	0.84	0.84	0.84	0.84
Nozzle diameter (inches):	0.226	0.226	0.221	0.226	0.256
Stack diameter (inches):	375.00	375.00	375.00	375.00	375.00
F _c :	1840	1840	1840	1840	1840

Method 5 Laboratory Data

Beaker Tare Weight (g):	53.3839	53.0772	51.8367	53.2265	52.6687
Beaker Final Weight (g):	53.3897	53.0864	51.8451	53.2357	52.6807
Filterable PM mass (probe) (g):	0.0058	0.0092	0.0084	0.0092	0.0120
Filter Tare Weight (g):	0.3561	0.3552	0.3556	0.3566	0.3555
Filter Final Weight (g):	0.3564	0.3558	0.3565	0.3573	0.3638
Filterable PM mass (filter) (g):	0.0003	0.0006	0.0009	0.0007	0.0083
Acetone Wash (ml):	76	56	68	74	89

Calculations

Maximum allowed blank correction (g):	0.00060	0.00044	0.00053	0.00058	0.00070
Acetone blank PM, if allowed (g):	0.00000	0.00000	0.00000	0.00000	0.00000
Blank corrected PM (g):	0.00610	0.00980	0.00930	0.00990	0.02030
Sample volume (dscf) (Vmstd):	36.231	89.390	86.582	90.135	35.006
Moisture volume (scf) (Vwstd):	6.747	15.989	15.993	16.512	5.946
Moisture Fraction Measured (BWSmsd):	0.157	0.152	0.156	0.155	0.145
Moisture Fraction @ Saturation (BWSsat):	0.150	0.149	0.152	0.152	0.130
Moisture Fraction (BWS):	0.150	0.149	0.152	0.152	0.130
Molecular weight (dry) (Md):	30.27	30.32	30.29	30.27	30.03
Molecular weight (actual/wet) (Ms):	28.34	28.45	28.37	28.37	28.28
Gas velocity (ft/sec) (Vs):	61.2	60.6	61.5	61.3	45.2
Stack area ft ² (As):	766.99	766.99	766.99	766.99	766.99
Gas flow (acfm) (Qa):	2,817,945	2,789,954	2,830,835	2,821,239	2,080,309
Gas flow (dscfm) (Qs):	2,088,848	2,076,490	2,095,293	2,088,501	1,597,465
% Isokinetic:	100.3	99.0	99.6	99.4	99.6
Filterable PM (mg/acm):	3.4124	2.2302	2.1754	2.2246	12.0639



EPA Method 5 : Determination of Particulate Emissions from Stationary Sources

Field Data

	Run #11	Run #12	Run #13	Run #14	Averages
Date:	2/21/2025	2/21/2025	2/21/2025	2/21/2025	
Start time:	12:24	14:04	15:30	17:22	
Stop time:	13:23	15:04	16:40	18:18	
Sample volume (ft ³):	35.374	39.812	36.417	38.498	49.796
Sampling time (minutes):	48	48	48	48	63
Stack temp. (°F):	124	125	129	129	127.6
Meter temp. (°F):	69	70	72	75	68.8
Barometric pressure (in Hg):	29.25	29.12	29.19	29.19	29.19
Static pressure (in H ₂ O):	-0.72	-0.75	-0.78	-0.78	-0.856
Stack pressure (in H ₂ O):	29.20	29.06	29.13	29.13	29.131
Oxygen (%dry):	9.3	7.9	7.0	7.1	7.46
Carbon Dioxide (%dry):	10.4	11.6	12.6	12.5	12.11
Moisture (gm):	117.3	138.7	145.0	150.6	183.64
ΔH (in H ₂ O):	1.817	2.154	2.042	2.208	2.147
√Δp (in H ₂ O):	0.7371	0.8255	0.9910	1.0297	0.944
DGM Y _c :	0.986	0.986	0.986	0.986	0.986
Pitot Tube Coefficient (C _p):	0.84	0.84	0.84	0.84	0.840
Nozzle diameter (inches):	0.255	0.256	0.226	0.226	0.236
Stack diameter (inches):	375.00	375.00	375.00	375.00	375.00
F _c :	1840	1840	1840	1840	1840

Method 5 Laboratory Data

Beaker Tare Weight (g):	52.4223	54.1080	51.5006	64.6670	
Beaker Final Weight (g):	52.4401	54.1305	51.5042	64.6694	
Filterable PM mass (probe) (g):	0.0178	0.0225	0.0036	0.0024	0.00828
Filter Tare Weight (g):	0.3521	0.3550	0.3545	0.3580	
Filter Final Weight (g):	0.3597	0.3644	0.3547	0.3584	
Filterable PM mass (filter) (g):	0.0076	0.0094	0.0002	0.0004	0.00219
Acetone Wash (ml):	76	72	53	78	

Blank Volume (ml)	200
Reagent Blank (g)	0.00000
Acetone Density g/l	0.7846
Max Ca%residue	0.0010%
Ca%residue	0.0000%

Calculations

Maximum allowed blank correction (g):	0.00060	0.00056	0.00042	0.00061	0.00055
Acetone blank PM, if allowed (g):	0.00000	0.00000	0.00000	0.00000	0.00000
Blank corrected PM (g):	0.02540	0.03190	0.00380	0.00280	0.01047
Sample volume (dscf) (Vmstd):	34.183	38.280	34.941	36.763	48.0241
Moisture volume (scf) (Vwstd):	5.531	6.540	6.837	7.101	8.6584
Moisture Fraction Measured (BWSmsd):	0.139	0.146	0.164	0.162	0.152
Moisture Fraction @ Saturation (BWSsat):	0.130	0.135	0.150	0.152	0.146
Moisture Fraction (BWS):	0.130	0.135	0.150	0.152	0.146
Molecular weight (dry) (Md):	30.03	30.18	30.29	30.29	30.236
Molecular weight (actual/wet) (Ms):	28.35	28.40	28.28	28.30	28.371
Gas velocity (ft/sec) (Vs):	44.4	49.9	60.1	62.5	57.19
Stack area ft ² (As):	766.99	766.99	766.99	766.99	766.99
Gas flow (acfm) (Qa):	2,045,067	2,295,603	2,768,050	2,876,222	2632051.8
Gas flow (dscfm) (Qs):	1,570,403	1,743,324	2,056,465	2,129,964	1964988.2
% Isokinetic:	99.1	99.4	99.1	100.2	99.5
Filterable PM (mg/acm):	15.4581	17.1757	2.2080	1.5427	

EXAMPLE CALCULATIONS

Plant/Unit:	La Cygne Unit 1
Project No.:	25-012
Run No.:	1
Test Type:	RCAM5

Meter Pressure (Pm), in. Hg

$$Pm = Pb + \frac{\Delta H}{13.6}$$

where,

Pb	29.23	= barometric pressure, in. Hg
ΔH	3.283	= pressure differential of orifice, in H ₂ O
Pm	29.47	= in. Hg

Absolute Stack Gas Pressure (Ps), in. Hg

$$Ps = Pb + \frac{Pg}{13.6}$$

where,

Pb	29.23	= barometric pressure, in. Hg
Pg	-0.75	= static pressure, in. H ₂ O
Ps	29.17	= in. Hg

Standard Meter Volume (Vmstd), dscf

$$Vmstd = \frac{17.647 \times Y \times Vm \times Pm}{Tm}$$

where,

Y	0.986	= meter correction factor
Vm	46.748	= meter volume, cf
Pm	29.47	= absolute meter pressure, in. Hg
Tm	518.8	= absolute meter temperature, °R
Vmstd	46.204	= dscf

Standard Wet Volume (Vwstd), scf

$$Vwstd = 0.04707 \times Vlc$$

where,

Vlc	164.2	= volume of H ₂ O collected, ml
Vwstd	7.742	= scf

Moisture Fraction (BWSsat), dimensionless (theoretical at saturated conditions)

$$BWSsat = \frac{10^{6.37 - \left(\frac{2,827}{Ts + 365} \right)}}{Ps}$$

where,

Ts	128.5	= stack temperature, °F
Ps	29.17	= absolute stack gas pressure, in. Hg
BWSsat	0.149	= dimensionless

Moisture Fraction (BWS), dimensionless (measured)

$$BWS = \frac{Vwstd}{(Vwstd + Vmstd)}$$

where,

Vwstd	7.742	= standard wet volume, scf
Vmstd	46.204	= standard meter volume, dscf
BWS	0.144	= dimensionless

Moisture Fraction (BWS), dimensionless

$$BWS = BWS_{msd} \text{ unless } BWS_{sat} < BWS_{msd}$$

where,

BWS _{sat}	0.149	= moisture fraction (theoretical at saturated conditions)
BWS _{msd}	0.144	= moisture fraction (measured)
BWS	0.144	

Molecular Weight (DRY) (Md), lb/lb-mole

$$Md = (0.44 \times \% CO_2) + (0.32 \times \% O_2) + (0.28 (100 - \% CO_2 - \% O_2))$$

where,

CO ₂	11.9	= carbon dioxide concentration, %
O ₂	7.6	= oxygen concentration, %
Md	30.21	= lb/lb mol

Molecular Weight (WET) (Ms), lb/lb-mole

$$Ms = Md (1 - BWS) + 18 (BWS)$$

where,

Md	30.21	= molecular weight (DRY), lb/lb mol
BWS	0.144	= moisture fraction, dimensionless
Ms	28.46	= lb/lb mol

Average Velocity (Vs), ft/sec

$$Vs = 85.49 \times Cp \times (\Delta P^{1/2})_{avg} \times \sqrt{\frac{Ts}{Ps \times Ms}}$$

where,

Cp	0.840	= pitot tube coefficient
$\Delta P^{1/2}$	0.799	= velocity head of stack gas, (in. H ₂ O) ^{1/2}
Ts	588.5	= absolute stack temperature, °R
Ps	29.17	= absolute stack gas pressure, in. Hg
Ms	28.46	= molecular weight of stack gas, lb/lb mol
Vs	48.3	= ft/sec

Average Stack Gas Flow at Stack Conditions (Qa), acfm

$$Qa = 60 \times Vs \times As$$

where,

Vs	48.3	= stack gas velocity, ft/sec
As	766.99	= cross-sectional area of stack, ft ²
Qa	2,224,248	= acfm

Average Stack Gas Flow at Standard Conditions (Qs), dscfm

$$Qs = 17.647 \times Qa \times (1 - BWS) \times \frac{Ps}{Ts}$$

where,

Qa	2,224,248	= average stack gas flow at stack conditions, acfm
BWS	0.144	= moisture fraction, dimensionless
Ps	29.17	= absolute stack gas pressure, in. Hg
Ts	588.5	= absolute stack temperature, °R
Qs	1,666,627	= dscfm

Dry Gas Meter Calibration Check (Yqa), dimensionless

$$Yqa = \frac{Y \cdot \left(\frac{\Theta}{Vm} \sqrt{\frac{0.0319 \times Tm \times 29}{\Delta H @ \times \left(Pb + \frac{\Delta H avg.}{13.6} \right) \times Md}} \sqrt{\Delta H avg.} \right)}{Y} \times 100$$

where,

Y	0.986	= meter correction factor, dimensionless
Θ	48	= run time, min.
Vm	46.748	= total meter volume, dcf
Tm	518.8	= absolute meter temperature, °R
ΔH@	1.949	= orifice meter calibration coefficient, in. H ₂ O
Pb	29.23	= barometric pressure, in. Hg
ΔH avg	3.283	= average pressure differential of orifice, in H ₂ O
Md	30.21	= molecular weight (DRY), lb/lb mol
(ΔH) ^{1/2}	1.812	= average square-root pressure differential of orifice, (in. H ₂ O) ^{1/2}
Yqa	0.8	= dimensionless

Volume of Nozzle (Vn), ft³

$$Vn = \frac{Ts}{Ps} \left(0.002669 \times Vlc + \frac{Vm \times Pm \times Y}{Tm} \right)$$

where,

Ts	588.5	= absolute stack temperature, °R
Ps	29.17	= absolute stack gas pressure, in. Hg
Vlc	164.2	= volume of H ₂ O collected, ml
Vm	46.748	= meter volume, cf
Pm	29.47	= absolute meter pressure, in. Hg
Y	0.986	= meter correction factor, unitless
Tm	518.8	= absolute meter temperature, °R
Vn	61.654	= volume of nozzle, ft ³

Isokinetic Sampling Rate (I), %

$$I = \left(\frac{Vn}{\theta \times 60 \times An \times Vs} \right) \times 100$$

where,

Vn	61.654	= nozzle volume, ft ³
θ	48.0	= run time, minutes
An	0.00045	= area of nozzle, ft ²
Vs	48.3	= average velocity, ft/sec
I	99.3	= %

Filterable PM Concentration (Cs), grain/dscf

$$Cs = \frac{Mn \times 0.0154}{Vmstd}$$

where,

Mn	5.5	= filterable PM mass, mg
Vmstd	46.204	= standard meter volume, dscf
Cs	0.0018	= grain/dscf

Filterable PM Concentration (C_{PM}), mg/dscm

$$C_{PM} = \frac{M_n \times 35.313}{V_{nstd}}$$

where,

M _n	5.5	= filterable PM mass, mg
V _{nstd}	46.204	= standard meter volume, dscf
C _{PM}	4.2	= mg/dscm

Filterable PM Emission Factor (EF_{PM CO2}), lb/MMBtu

$$EF_{PM\ CO_2} = \frac{Cs \times Fc}{7.0E+03} \times \frac{100}{CO_2}$$

where,

C _s	0.0018	= filterable PM concentration, grain/dscf
F _c	1,840	= carbon dioxide based fuel factor, dscf/MMBtu
CO ₂	11.9	= carbon dioxide concentration, %
EF _{PM CO2}	0.004	= lb/MMBtu

REFERENCE METHOD DATA

METHOD 3-3A DATA

Plant/Unit: La Cygne Unit 1
 Project No.: 25-012

Molecular Weight

Method 3-3A Tedlar Bag

Analyzer Calibration

Calibration Date/Time		Analyzer Make/Model/SN			
2/19/2025	8:00	Teledyne	T802	156	
		O ₂ Data		CO ₂ Data	
	Cylinder ID	Cylinder Concentration (%)	Analyzer Response (%)	Cylinder Concentration (%)	Analyzer Response (%)
Zero Gas	UHP Nitrogen	0.00	0.00	0.00	0.00
Span Cal Gas	DT0031874	20.00	20.01	20.10	20.10
Mid Cal Gas	CC199274	9.98	10.01	9.61	9.60
	Required Accuracy	± 0.4% of cylinder concentration		± 0.4% of cylinder concentration	

Tedlar Bag Analysis

Tedlar Bag ID	Start	End	% O ₂	% CO ₂
Run 1	10:34	11:37	7.55	11.92

Tedlar Bag ID	Start	End	% O ₂	% CO ₂
Run 2	12:00	13:03	7.09	12.51

Tedlar Bag ID	Start	End	% O ₂	% CO ₂
Run 3	13:12	14:15	7.01	12.57

Tedlar Bag ID	Start	End	% O ₂	% CO ₂
Run 4	14:48	15:51	7.04	12.53

Tedlar Bag ID	Start	End	% O ₂	% CO ₂
Run 5	16:00	17:03	7.09	12.48

Tedlar Bag ID	Start	End	% O ₂	% CO ₂
Run 6	17:07	18:10	7.10	12.40

Plant/Unit: La Cygne Unit 1
 Project No.: 25-012

Molecular Weight

Method 3-3A Tedlar Bag

Analyzer Calibration

Calibration Date/Time		Analyzer Make/Model/SN			
2/20/2025	8:35	Teledyne	T802	156	
		O ₂ Data		CO ₂ Data	
	Cylinder ID	Cylinder Concentration (%)	Analyzer Response (%)	Cylinder Concentration (%)	Analyzer Response (%)
Zero Gas	UHP Nitrogen	0.00	0.00	0.00	0.00
Span Cal Gas	DT0031874	20.00	20.00	20.10	20.12
Mid Cal Gas	CC199274	9.98	10.01	9.61	9.62
	Required Accuracy	± 0.4% of cylinder concentration		± 0.4% of cylinder concentration	

Tedlar Bag Analysis

Tedlar Bag ID	Start	End	% O ₂	% CO ₂
Run 7	13:00	15:15	6.81	12.79

Tedlar Bag ID	Start	End	% O ₂	% CO ₂
Run 8	15:26	17:41	6.97	12.56

Tedlar Bag ID	Start	End	% O ₂	% CO ₂
Run 9	17:51	20:06	7.14	12.39

Plant/Unit: La Cygne Unit 1
 Project No.: 25-012

Molecular Weight

Method 3-3A Tedlar Bag

Analyzer Calibration

Calibration Date/Time		Analyzer Make/Model/SN			
2/21/2025	8:15	Teledyne	T802	156	
		O ₂ Data		CO ₂ Data	
	Cylinder ID	Cylinder Concentration (%)	Analyzer Response (%)	Cylinder Concentration (%)	Analyzer Response (%)
Zero Gas	UHP Nitrogen	0.00	0.00	0.00	0.00
Span Cal Gas	DT0031874	20.00	20.01	20.10	20.11
Mid Cal Gas	CC199274	9.98	10.00	9.61	9.64
	Required Accuracy	± 0.4% of cylinder concentration		± 0.4% of cylinder concentration	

Tedlar Bag Analysis

Tedlar Bag ID	Start	End	% O ₂	% CO ₂
Run 10	10:55	11:53	9.28	10.35

Tedlar Bag ID	Start	End	% O ₂	% CO ₂
Run 11	12:24	13:23	9.32	10.35

Tedlar Bag ID	Start	End	% O ₂	% CO ₂
Run 12	14:04	15:04	7.90	11.63

Tedlar Bag ID	Start	End	% O ₂	% CO ₂
Run 13	15:30	16:40	6.98	12.56

Tedlar Bag ID	Start	End	% O ₂	% CO ₂
Run 14	17:22	18:18	7.11	12.54

PARTICULATE MATTER FIELD DATA

Evergy - Field Data

Plant, Unit: La Cygne Unit 1
 Run #: 1
 Date: 2/19/2025
 Operator: SB
 Test Type: RCA M5
 Run Time: 48 minutes
 Start/End: 10:34 11:37
 Estimated Final
 O₂, %: 6.00 7.55
 CO₂, %: 14.00 11.92

Barometric Pressure: 29.23 inHG
 Static Pressure: -0.75 inH₂O
 Stack Diameter/Area: 375 in. / 766.99 sqft
 itot Tube Coefficient: 0.84
 DGM Coefficient (Y): 0.986
 Delta H @: 1.949
 Meter Box #: 1806015
 Nozzle Diameter: 0.286
 Assumed % Moisture: 16
 Final % Moisture: 14.4 Actual

Leak Check, I: 0.000 @ 15 inHG
 Leak Check, F: 0.000 @ 5 inHG
 Pitot Leak Check, I: Pass
 Pitot Leak Check, F: Pass
 Probe ID: Glass M5 11ft A
 Pitot ID: A8481
 Filter ID Number: 5B333
 Project Number: 25-012

Point	Sample Time	DGM (ft³)	Delta H	Vacuum (in. Hg)	Delta P	Stack Temp. °F	Probe Temp °F	Filter Temp. °F	CPM Filter Temp. °F	Cold Box Temp	DGM Temp. °F			
A-1	10:34	596.170	2.80	5	0.56	128	250	250	NA	48	54			
2	10:38	599.850	2.80	5	0.55	128	250	250	NA	46	54			
3	10:42	603.370	2.70	5	0.54	129	250	250	NA	46	55			
B-1	10:51	606.960	2.90	5	0.57	129	250	250	NA	46	55			
2	10:55	610.690	3.20	5	0.62	128	250	250	NA	48	57			
3	10:59	614.490	3.00	5	0.58	128	250	250	NA	48	58			
C-1	11:08	618.220	3.20	5	0.63	128	250	250	NA	52	59			
2	11:12	622.040	3.40	5	0.67	128	250	250	NA	55	61			
3	11:16	625.980	3.80	5	0.74	128	250	250	NA	55	61			
D-1	11:25	630.180	3.70	5	0.72	129	250	250	NA	54	63			
2	11:29	634.490	4.10	5	0.79	128	250	250	NA	55	64			
3	11:33	638.650	3.80	5	0.73	128	250	250	NA	54	65			
End	11:37	642.918												
2988.2		Total ft³:	Avg:	High:	Avg Sq. Rt.:		High:	Low:	High:	Low:	High:	Low:	High:	Avg:
		46.748	3.28	5	0.7994	129	128	250	250	250	250			55

Initial Wt: 2988.2
 Final Wt: 3152.4
 Weight Gain: 164.2

DGM Calibration Check (Yqa): 0.8
 (Yqa must be ± 5 to be valid)

K-Factor: 5.15
 % ISO: 99.3

Evergy - Field Data

Plant, Unit: La Cygne Unit 1
 Run #: 2
 Date: 2/19/2025
 Operator: SB
 Test Type: RCA M5
 Run Time: 48 minutes
 Start/End: 12:00 13:03
 Estimated Final
 O₂, %: 6.00 7.09
 CO₂, %: 14.00 12.51

Barometric Pressure: 29.21 inHG
 Static Pressure: -0.89 inH₂O
 Stack Diameter/Area: 375 in. / 766.99 sqft
 itot Tube Coefficient: 0.84
 DGM Coefficient (Y): 0.986
 Delta H @: 1.949
 Meter Box #: 1806015
 Nozzle Diameter: 0.226
 Assumed % Moisture: 16
 Final % Moisture: 14.5 Actual

Leak Check, I: 0.000 @ 15 inHG
 Leak Check, F: 0.000 @ 5 inHG
 Pitot Leak Check, I: Pass
 Pitot Leak Check, F: Pass
 Probe ID: Glass M5 11ft B
 Pitot ID: A8482
 Filter ID Number: 5B334
 Project Number: 25-012

Point	Sample Time	DGM (ft³)	Delta H	Vacuum (in. Hg)	Delta P	Stack Temp. °F	Probe Temp °F	Filter Temp. °F	CPM Filter Temp. °F	Cord Box Temp	DGM Temp. °F			
A-1	12:00	643.254	2.00	5	0.98	128	250	250	NA	50	60			
2	12:04	646.290	2.00	5	1.00	128	250	250	NA	43	60			
3	12:08	649.310	2.20	5	1.10	128	250	250	NA	43	60			
B-1	12:17	652.615	2.00	5	1.00	128	250	250	NA	43	60			
2	12:21	655.650	2.40	5	1.20	128	250	250	NA	46	61			
3	12:25	658.860	2.00	5	0.98	128	250	250	NA	46	62			
C-1	12:34	661.970	2.40	5	1.20	129	250	250	NA	48	63			
2	12:38	665.270	2.20	5	1.10	128	250	250	NA	49	64			
3	12:42	668.450	2.00	5	1.00	128	250	250	NA	50	64			
D-1	12:51	671.500	2.40	5	1.20	128	252	250	NA	51	65			
2	12:55	674.760	2.20	5	1.10	129	251	250	NA	51	66			
3	12:59	678.010	2.00	5	1.00	128	250	252	NA	52	66			
End	13:03	681.030		5										
3013.5		Total ft³:	Avg:	High:	Avg Sq. Rt.:		High:	Low:	High:	Low:	High:	Low:	High:	Avg:
		37.776	2.15	5	1.0344	129	128	252	250	252	250		52	63

Initial Wt: 3013.5
 Final Wt: 3146.8
 Weight Gain: 133.3

DGM Calibration Check (Yqa): 0.2
 (Yqa must be ± 5 to be valid)

K-Factor: 2.02
 % ISO: 98.6

Evergy - Field Data

Plant, Unit: La Cygne Unit 1
 Run #: 3
 Date: 2/19/2025
 Operator: SB
 Test Type: RCA M5
 Run Time: 48 minutes
 Start/End: 13:12 14:15
 Estimated Final
 O₂, %: 6.00 7.01
 CO₂, %: 14.00 12.57

Barometric Pressure: 29.19 inHG
 Static Pressure: -0.95 inH₂O
 Stack Diameter/Area: 375 in. / 766.99 sqft
 itot Tube Coefficient: 0.84
 DGM Coefficient (Y): 0.986
 Delta H @: 1.949
 Meter Box #: 1806015
 Nozzle Diameter: 0.226
 Assumed % Moisture: 16
 Final % Moisture: 15 Saturated

Leak Check, I: 0.000 @ 15 inHG
 Leak Check, F: 0.000 @ 5 inHG
 Pitot Leak Check, I: Pass
 Pitot Leak Check, F: Pass
 Probe ID: Glass M5 11ft A
 Pitot ID: A8481
 Filter ID Number: 5B335
 Project Number: 25-012

Point	Sample Time	DGM (ft³)	Delta H	Vacuum (in. Hg)	Delta P	Stack Temp. °F	Probe Temp °F	Filter Temp. °F	CPM Filter Temp. °F	Cord Box Temp	DGM Temp. °F			
A-1	13:12	681.223	2.20	5	1.10	128	251	251	NA	53	63			
2	13:16	684.460	2.00	5	1.00	128	248	250	NA	47	64			
3	13:20	687.580	2.00	5	0.98	128	250	250	NA	47	64			
B-1	13:29	690.510	2.20	5	1.10	128	251	251	NA	48	64			
2	13:33	693.600	2.00	5	0.99	129	250	250	NA	50	65			
3	13:37	696.730	2.00	5	0.97	129	251	250	NA	53	66			
C-1	13:46	699.740	2.20	5	1.10	128	251	250	NA	55	66			
2	13:50	702.840	2.00	5	1.00	128	321	320	NA	57	67			
3	13:54	705.880	2.00	5	0.97	129	320	322	NA	59	68			
D-1	14:03	708.910	2.00	5	1.00	128	321	322	NA	59	68			
2	14:07	711.945	2.00	5	0.98	128	321	321	NA	63	69			
3	14:11	714.990	2.00	5	0.96	128	321	321	NA	60	69			
End	14:15	718.028												
3011.3		Total ft³:	Avg:	High:	Avg Sq. Rt.:		High:	Low:	High:	Low:	High:	Low:	High:	Avg:
		36.805	2.05	5	1.0059	129	128	321	248	322	250			63

Initial Wt: 3011.3
 Final Wt: 3152.2
 Weight Gain: 140.9

DGM Calibration Check (Yqa): -0.4
 (Yqa must be ± 5 to be valid)

K-Factor: 2.03
 % ISO: 99.1

Evergy - Field Data

Plant, Unit: La Cygne Unit 1
 Run #: 4
 Date: 2/19/2025
 Operator: SB
 Test Type: RCA M5
 Run Time: 48 minutes
 Start/End: 14:48 15:51
 Estimated Final
 O₂, %: 6.00 7.04
 CO₂, %: 14.00 12.53

Barometric Pressure: 29.16 inHG
 Static Pressure: -0.98 inH₂O
 Stack Diameter/Area: 375 in. / 766.99 sqft
 itot Tube Coefficient: 0.84
 DGM Coefficient (Y): 0.986
 Delta H @: 1.949
 Meter Box #: 1806015
 Nozzle Diameter: 0.226
 Assumed % Moisture: 16
 Final % Moisture: 15 Saturated

Leak Check, I: 0.000 @ 15 inHG
 Leak Check, F: 0.000 @ 5 inHG
 Pitot Leak Check, I: Pass
 Pitot Leak Check, F: Pass
 Probe ID: Glass M5 11ft B
 Pitot ID: A8482
 Filter ID Number: 5B440
 Project Number: 25-012

Point	Sample Time	DGM (ft³)	Delta H	Vacuum (in. Hg)	Delta P	Stack Temp. °F	Probe Temp °F	Filter Temp. °F	CPM Filter Temp. °F	Cord Box Temp	DGM Temp. °F			
A-1	14:48	719.267	2.20	5	1.10	128	318	320	NA	48	63			
2	14:52	722.530	2.00	5	1.00	128	322	321	NA	47	64			
3	14:56	725.650	2.00	5	0.98	129	319	320	NA	48	66			
B-1	15:05	728.560	2.30	5	1.10	128	321	319	NA	50	66			
2	15:09	731.840	2.00	5	0.99	128	318	320	NA	52	67			
3	15:13	734.870	2.00	5	0.98	129	321	321	NA	54	68			
C-1	15:22	737.870	2.30	5	1.10	129	321	321	NA	54	69			
2	15:26	741.110	2.00	5	0.99	128	320	320	NA	51	69			
3	15:30	744.210	1.90	5	0.94	128	321	320	NA	50	70			
D-1	15:39	747.340	2.30	5	1.10	128	321	320	NA	50	71			
2	15:43	750.480	2.00	5	0.99	128	321	321	NA	48	71			
3	15:47	753.530	2.00	5	0.97	128	320	321	NA	49	72			
End	15:51	756.695												
3022.3		Total ft³:	Avg:	High:	Avg Sq. Rt.:		High:	Low:	High:	Low:	High:	Low:	High:	Avg:
		37.428	2.08	5	1.0095	129	128	322	318	321	319		54	68

Initial Wt: 3022.3
 Final Wt: 3163
 Weight Gain: 140.7

DGM Calibration Check (Yqa): 0.3
 (Yqa must be ± 5 to be valid)

K-Factor: 2.04
 % ISO: 99.9

Evergy - Field Data

Plant, Unit: La Cygne Unit 1
Run #: 5
Date: 2/19/2025
Operator: SB
Test Type: RCA M5
Run Time: 48 minutes
Start/End: 16:00 17:03
Estimated Final
O₂, %: 6.00 7.09
CO₂, %: 14.00 12.48

Barometric Pressure: 29.15 inHG
Static Pressure: -0.98 inH₂O
Stack Diameter/Area: 375 in. / 766.99 sqft
itot Tube Coefficient: 0.84
DGM Coefficient (Y): 0.986
Delta H @: 1.949
Meter Box #: 1806015
Nozzle Diameter: 0.226
Assumed % Moisture: 16
Final % Moisture: 15 Saturated

Leak Check, I: 0.000 @ 15 inHG
Leak Check, F: 0.000 @ 5 inHG
Pitot Leak Check, I: Pass
Pitot Leak Check, F: Pass
Probe ID: Glass M5 11ft A
Pitot ID: A8481
Filter ID Number: 5B441
Project Number: 25-012

Point	Sample Time	DGM (ft³)	Delta H	Vacuum (in. Hg)	Delta P	Stack Temp. °F	Probe Temp °F	Filter Temp. °F	CPM Filter Temp. °F	Cord Box Temp	DGM Temp. °F			
A-1	16:00	757.300	2.10	5	1.00	128	321	320	NA	52	67			
2	16:04	760.280	2.00	5	0.99	128	321	324	NA	43	67			
3	16:08	763.350	2.00	5	0.98	128	319	320	NA	44	68			
B-1	16:17	766.405	2.30	5	1.10	128	321	320	NA	45	68			
2	16:21	769.680	2.00	5	0.98	128	321	320	NA	46	69			
3	16:25	772.710	2.00	5	0.97	128	319	320	NA	46	69			
C-1	16:34	775.720	2.10	5	1.00	128	320	320	NA	47	70			
2	16:38	778.860	2.00	5	0.99	128	320	321	NA	47	70			
3	16:42	781.870	2.00	5	0.97	128	319	320	NA	48	71			
D-1	16:51	784.960	2.30	5	1.10	129	321	320	NA	48	71			
2	16:55	788.265	2.10	5	1.00	129	320	320	NA	48	72			
3	16:59	791.350	2.00	5	0.98	129	319	320	NA	49	72			
End	17:03	794.490												
3000		Total ft³:	Avg:	High:	Avg Sq. Rt.:		High:	Low:	High:	Low:	High:	Low:	High:	Avg:
		37.190	2.08	5	1.0023	129	128	321	319	324	320			52

Initial Wt: 3000
Final Wt: 3142.5
Weight Gain: 142.5

DGM Calibration Check (Yqa): -0.4
(Yqa must be ± 5 to be valid)

K-Factor: 2.05
% ISO: 99.9

Evergy - Field Data

Plant, Unit: La Cygne Unit 1
Run #: 6
Date: 2/19/2025
Operator: SB
Test Type: RCA M5
Run Time: 48 minutes
Start/End: 17:07 18:10
Estimated Final
O₂, %: 6.00 7.10
CO₂, %: 14.00 12.40

Barometric Pressure: 29.15 inHG
Static Pressure: -0.99 inH₂O
Stack Diameter/Area: 375 in. / 766.99 sqft
itot Tube Coefficient: 0.84
DGM Coefficient (Y): 0.986
Delta H @: 1.949
Meter Box #: 1806015
Nozzle Diameter: 0.226
Assumed % Moisture: 16
Final % Moisture: 15 Saturated

Leak Check, I: 0.000 @ 15 inHG
Leak Check, F: 0.000 @ 5 inHG
Pitot Leak Check, I: Pass
Pitot Leak Check, F: Pass
Probe ID: Glass M5 11ft B
Pitot ID: A8482
Filter ID Number: 5B442
Project Number: 25-012

Point	Sample Time	DGM (ft³)	Delta H	Vacuum (in. Hg)	Delta P	Stack Temp. °F	Probe Temp °F	Filter Temp. °F	CPM Filter Temp. °F	Cold Box Temp	DGM Temp. °F			
A-1	17:07	794.670	2.30	5	1.10	129	321	320	NA	45	68			
2	17:11	798.030	2.10	5	1.00	129	320	323	NA	48	68			
3	17:15	801.150	1.80	5	0.90	129	320	321	NA	49	69			
B-1	17:24	803.990	2.30	5	1.10	129	319	320	NA	50	69			
2	17:28	807.210	2.10	5	1.00	129	319	320	NA	52	70			
3	17:32	810.360	2.00	5	0.99	129	319	320	NA	54	70			
C-1	17:41	813.380	2.30	5	1.10	128	319	320	NA	56	71			
2	17:45	816.670	2.10	5	1.00	129	321	320	NA	58	71			
3	17:49	819.790	2.00	5	0.97	129	320	320	NA	60	72			
D-1	17:58	822.870	2.30	5	1.10	129	320	320	NA	56	72			
2	18:02	826.140	2.10	5	1.00	129	320	320	NA	52	72			
3	18:06	829.140	2.00	5	0.97	129	319	320	NA	50	73			
End	18:10	832.358												
3073.3		Total ft³:	Avg:	High:	Avg Sq. Rt.:		High:	Low:	High:	Low:	High:	Low:	High:	Avg:
		37.688	2.12	5	1.0091	129	128	321	319	323	320		60	70

Initial Wt: 3073.3
Final Wt: 3216.4
Weight Gain: 143.1

DGM Calibration Check (Yqa): -0.1
(Yqa must be ± 5 to be valid)

K-Factor: 2.05
% ISO: 100.3

Evergy - Field Data

Plant, Unit: La Cygne Unit 1
 Run #: 7
 Date: 2/20/2025
 Operator: SB
 Test Type: RCA M5
 Run Time: 120 minutes
 Start/End: 13:00 15:15
 Estimated Final
 O₂, %: 6.00 6.81
 CO₂, %: 14.00 12.79

Barometric Pressure: 29.24 inHG
 Static Pressure: -0.85 inH₂O
 Stack Diameter/Area: 375 in. / 766.99 sqft
 itot Tube Coefficient: 0.84
 DGM Coefficient (Y): 0.986
 Delta H @: 1.949
 Meter Box #: 1806015
 Nozzle Diameter: 0.226
 Assumed % Moisture: 16
 Final % Moisture: 14.9 Saturated

Leak Check, I: 0.000 @ 15 inHG
 Leak Check, F: 0.000 @ 5 inHG
 Pitot Leak Check, I: Pass
 Pitot Leak Check, F: Pass
 Probe ID: Glass M5 11ft A
 Pitot ID: A8481
 Filter ID Number: 5B443
 Project Number: 25-012

Point	Sample Time	DGM (ft³)	Delta H	Vacuum (in. Hg)	Delta P	Stack Temp. °F	Probe Temp °F	Filter Temp. °F	CPM Filter Temp. °F	Cord Box Temp	DGM Temp. °F				
A-1	13:00	840.748	2.20	5	1.10	128	322	326	66	56	60				
1	13:05	844.570	2.20	5	1.10	128	321	320	65	55	61				
2	13:10	848.510	2.00	5	1.00	128	322	320	65	51	63				
2	13:15	852.250	2.00	5	0.99	129	320	320	65	49	64				
3	13:20	855.990	2.00	5	0.97	128	320	320	69	48	65				
3	13:25	859.800	2.00	5	0.97	128	320	320	82	48	66				
B-1	13:35	863.530	2.30	5	1.10	128	320	320	84	50	67				
1	13:40	867.510	2.30	5	1.10	128	320	320	65	56	68				
2	13:45	871.570	2.00	5	0.99	128	321	320	65	53	69				
2	13:50	875.410	2.00	5	0.98	128	321	320	75	54	70				
3	13:55	879.240	2.00	5	0.96	128	321	320	84	56	70				
3	14:00	882.990	1.90	5	0.94	128	320	320	75	57	71				
C-1	14:10	886.780	2.30	5	1.10	128	320	320	66	56	71				
1	14:15	890.820	2.30	5	1.10	129	321	320	69	60	72				
2	14:20	894.940	2.00	5	0.96	128	319	319	65	56	73				
2	14:25	898.750	2.00	5	0.95	129	320	320	65	54	74				
3	14:30	902.570	1.90	5	0.90	129	318	320	76	56	73				
3	14:35	906.240	1.90	5	0.93	129	318	320	78	53	73				
D-1	14:45	909.940	2.30	5	1.10	129	321	320	67	53	73				
1	14:50	914.110	2.50	5	1.20	129	318	320	65	56	74				
2	14:55	918.320	2.10	5	1.00	129	319	320	78	57	74				
2	15:00	922.220	2.10	5	1.00	129	321	320	80	58	74				
3	15:05	926.060	1.80	5	0.88	129	321	319	69	60	74				
3	15:10	929.690	1.80	5	0.85	129	320	320	65	62	74				
End	15:15	933.332													
2550.6		Total ft³:	Avg:	High:	Avg Sq. Rt.:	High:	Low:	High:	Low:	High:	Low:	High:	Low:	High:	Avg:
		92.584	2.08	5	1.0027	129	128	322	318	326	319	84	65	62	70

Initial Wt: 2550.6
 Final Wt: 2889.7
 Weight Gain: 339.1

DGM Calibration Check (Yqa): -0.7
 (Yqa must be ± 5 to be valid)

K-Factor: 2.05
 % ISO: 99.0

Evergy - Field Data

Plant, Unit: La Cygne Unit 1
 Run #: 8
 Date: 2/20/2025
 Operator: SB
 Test Type: RCA M5
 Run Time: 120 minutes
 Start/End: 15:26 17:41

	Estimated	Final
O ₂ , %:	6.00	6.97
CO ₂ , %:	14.00	12.56

Barometric Pressure: 29.19 inHG
 Static Pressure: -0.95 inH₂O
 Stack Diameter/Area: 375 in. / 766.99 sqft
 itot Tube Coefficient: 0.84
 DGM Coefficient (Y): 0.986
 Delta H @: 1.949
 Meter Box #: 1806015
 Nozzle Diameter: 0.221
 Assumed % Moisture: 16
 Final % Moisture: 15.2 Saturated

Leak Check, I: 0.000 @ 15 inHG
 Leak Check, F: 0.000 @ 5 inHG
 Pitot Leak Check, I: Pass
 Pitot Leak Check, F: Pass
 Probe ID: Glass M5 11ft B
 Pitot ID: A8482
 Filter ID Number: 5B444
 Project Number: 25-012

Point	Sample Time	DGM (ft³)	Delta H	Vacuum (in. Hg)	Delta P	Stack Temp. °F	Probe Temp °F	Filter Temp. °F	CPM Filter Temp. °F	Cold Box Temp	DGM Temp. °F				
A-1	15:26	934.071	2.10	5	1.10	129	321	321	65	47	68				
1	15:31	938.030	2.30	5	1.20	129	319	320	75	55	69				
2	15:36	942.090	1.90	5	1.00	129	320	321	85	58	69				
2	15:41	945.780	1.90	5	0.99	129	321	320	73	60	70				
3	15:46	949.480	1.80	5	0.98	129	320	320	67	62	70				
3	15:51	953.090	1.80	5	0.98	129	319	320	85	59	71				
B-1	16:01	956.725	2.10	5	1.10	129	321	320	72	54	71				
1	16:06	960.670	2.30	5	1.20	129	320	320	65	53	72				
2	16:11	964.670	1.90	5	1.00	129	320	320	76	54	73				
2	16:16	968.410	1.90	5	1.00	129	320	320	85	55	73				
3	16:21	972.150	1.80	5	0.96	129	321	320	84	57	74				
3	16:26	975.780	1.80	5	0.95	129	321	320	69	58	74				
C-1	16:36	979.340	2.30	5	1.20	129	321	320	85	59	74				
1	16:41	983.560	2.10	5	1.10	129	321	320	85	59	75				
2	16:46	987.610	1.90	5	1.00	129	321	320	72	61	76				
2	16:51	991.170	1.90	5	0.99	129	319	320	85	62	76				
3	16:56	994.860	1.80	5	0.96	129	318	320	78	61	76				
3	17:01	998.450	2.10	5	1.10	129	320	320	68	59	76				
D-1	17:11	1002.310	2.30	5	1.20	129	320	320	85	58	76				
1	17:16	1006.470	1.90	5	0.98	129	320	320	77	59	77				
2	17:21	1010.180	1.90	5	0.98	129	320	320	68	60	77				
2	17:26	1013.880	1.80	5	0.95	129	320	320	85	60	77				
3	17:31	1017.510	1.70	5	0.90	129	320	320	76	60	76				
3	17:36	1021.030	1.80	5	0.93	129	320	320	69	61	77				
End	17:41	1024.590													
2571.1	Total ft³:		Avg:	High:	Avg Sq. Rt.:	High:	Low:	High:	Low:	High:	Low:	High:	Low:	High:	Avg:
	90.519		1.96	5	1.0145	129	129	321	318	321	320	85	65	62	74

Initial Wt: 2571.1
 Final Wt: 2910.3
 Weight Gain: 339.2

DGM Calibration Check (Yqa): -0.6
 (Yqa must be ± 5 to be valid)

K-Factor: 1.89
 % ISO: 99.6

Evergy - Field Data

Plant, Unit: La Cygne Unit 1
 Run #: 9
 Date: 2/20/2025
 Operator: SB
 Test Type: RCA M5
 Run Time: 120 minutes
 Start/End: 17:51 20:06
 Estimated Final
 O₂, %: 7.00 7.14
 CO₂, %: 14.00 12.39

Barometric Pressure: 29.19 inHG
 Static Pressure: -0.90 inH₂O
 Stack Diameter/Area: 375 in. / 766.99 sqft
 itot Tube Coefficient: 0.84
 DGM Coefficient (Y): 0.986
 Delta H @: 1.949
 Meter Box #: 1806015
 Nozzle Diameter: 0.226
 Assumed % Moisture: 16
 Final % Moisture: 15.2 Saturated

Leak Check, I: 0.000 @ 15 inHG
 Leak Check, F: 0.000 @ 5 inHG
 Pitot Leak Check, I: Pass
 Pitot Leak Check, F: Pass
 Probe ID: Glass M5 11ft A
 Pitot ID: A8481
 Filter ID Number: 5B445
 Project Number: 25-012

Point	Sample Time	DGM (ft³)	Delta H	Vacuum (in. Hg)	Delta P	Stack Temp. °F	Probe Temp °F	Filter Temp. °F	CPM Filter Temp. °F	Cord Box Temp	DGM Temp. °F				
A-1	17:51	25.922	2.30	5	1.10	129	321	320	66	54	71				
1	17:56	29.920	2.30	5	1.10	129	320	321	65	51	70				
2	18:01	33.850	2.10	5	1.00	129	321	320	67	50	71				
2	18:06	37.790	2.00	5	0.99	129	320	320	79	49	71				
3	18:11	41.580	2.00	5	0.98	129	320	320	73	50	72				
3	18:16	45.420	2.00	5	0.97	129	320	320	66	50	72				
B-1	18:26	49.230	2.30	5	1.10	129	320	320	65	52	73				
1	18:31	53.320	2.30	5	1.10	129	322	320	78	53	74				
2	18:36	57.440	2.10	5	1.00	129	320	320	85	54	75				
2	18:41	61.350	2.10	5	0.99	129	321	320	84	56	75				
3	18:46	65.320	2.00	5	0.97	129	320	320	74	57	75				
3	18:51	69.110	2.00	5	0.96	129	321	320	72	58	75				
C-1	19:01	72.910	2.50	5	1.20	129	320	320	71	58	76				
1	19:06	77.220	2.30	5	1.10	129	320	320	73	58	76				
2	19:11	81.380	2.00	5	0.98	129	320	320	74	59	77				
2	19:16	85.220	2.00	5	0.97	129	321	320	72	60	77				
3	19:21	89.035	2.00	5	0.95	129	322	320	70	60	77				
3	19:26	92.850	2.00	5	0.95	129	319	320	68	61	77				
D-1	19:36	96.670	2.30	5	1.10	129	319	320	68	62	77				
1	19:41	100.810	2.50	5	1.20	129	320	320	68	57	78				
2	19:46	105.130	2.10	5	1.00	129	319	320	69	58	78				
2	19:51	108.950	2.10	5	1.00	129	320	320	68	54	78				
3	19:56	112.850	2.00	5	0.94	129	320	320	67	54	78				
3	20:01	116.710	1.90	5	0.92	129	320	320	68	54	78				
End	20:06	120.365													
2589.7		Total ft³:	Avg:	High:	Avg Sq. Rt.:	High:	Low:	High:	Low:	High:	Low:	High:	Low:	High:	Avg:
		94.443	2.13	5	1.0111	129	129	322	319	321	320	85	65	62	75

Initial Wt: 2589.7
 Final Wt: 2939.9
 Weight Gain: 350.2

DGM Calibration Check (Yqa): -0.6
 (Yqa must be ± 5 to be valid)

K-Factor: 2.07
 % ISO: 99.4

Evergy - Field Data

Plant, Unit: La Cygne Unit 1
Run #: 10
Date: 2/21/2025
Operator: SB
Test Type: RCA M5
Run Time: 48 minutes
Start/End: 10:55 11:53
Estimated Final
O₂, %: 9.00 9.28
CO₂, %: 10.50 10.35

Barometric Pressure: 29.25 inHG
Static Pressure: -0.72 inH₂O
Stack Diameter/Area: 375 in. / 766.99 sqft
itot Tube Coefficient: 0.84
DGM Coefficient (Y): 0.986
Delta H @: 1.949
Meter Box #: 1806015
Nozzle Diameter: 0.256
Assumed % Moisture: 16
Final % Moisture: 13 Saturated

Leak Check, I: 0.000 @ 15 inHG
Leak Check, F: 0.000 @ 5 inHG
Pitot Leak Check, I: Pass
Pitot Leak Check, F: Pass
Probe ID: Glass M5 11ft A
Pitot ID: A8481
Filter ID Number: 5B446
Project Number: 25-012

Point	Sample Time	DGM (ft ³)	Delta H	Vacuum (in. Hg)	Delta P	Stack Temp. °F	Probe Temp °F	Filter Temp. °F	CPM Filter Temp. °F	Cord Box Temp	DGM Temp. °F		
A-1	10:55	121.015	2.00	5	0.60	124	319	320	NA	59	60		
2	10:59	124.185	2.00	5	0.58	123	320	320	NA	54	60		
3	11:03	127.180	1.70	5	0.51	123	321	319	NA	51	61		
B-1	11:10	130.075	2.00	5	0.59	124	320	321	NA	49	61		
2	11:14	133.170	2.00	5	0.59	124	320	320	NA	49	62		
3	11:18	136.125	1.70	5	0.50	124	319	319	NA	49	63		
C-1	11:26	138.955	2.00	5	0.60	124	321	320	NA	49	64		
2	11:30	142.020	2.00	5	0.58	124	319	321	NA	49	65		
3	11:34	145.115	1.70	5	0.49	124	321	320	NA	49	66		
D-1	11:41	147.910	2.10	5	0.61	124	321	321	NA	50	67		
2	11:45	151.06	2.00	5	0.59	124	318	320	NA	51	68		
3	11:49	154.155	1.70	5	0.50	124	321	320	NA	52	69		
End	11:53	156.873											
2947		Total ft ³ :	Avg:	High:	Avg Sq. Rt.:	High:	Low:	High:	Low:	High:	Low:	High:	Avg:
		35.858	1.91	5	0.7488	124	123	321	318	321	319		59

Initial Wt: 2947
Final Wt: 3073.1
Weight Gain: 126.1

DGM Calibration Check (Yqa): 0.5
(Yqa must be ± 5 to be valid)

K-Factor: 3.36
% ISO: 99.6

Evergy - Field Data

Plant, Unit: La Cygne Unit 1
Run #: 11
Date: 2/21/2025
Operator: BJ
Test Type: RCA M5
Run Time: 48 minutes
Start/End: 12:24 13:23
Estimated Final
O₂, %: 9.00 9.32
CO₂, %: 10.50 10.35

Barometric Pressure: 29.25 inHG
Static Pressure: -0.72 inH₂O
Stack Diameter/Area: 375 in. / 766.99 sqft
itot Tube Coefficient: 0.84
DGM Coefficient (Y): 0.986
Delta H @: 1.949
Meter Box #: 1806015
Nozzle Diameter: 0.255
Assumed % Moisture: 16
Final % Moisture: 13 Saturated

Leak Check, I: 0.000 @ 15 inHG
Leak Check, F: 0.000 @ 5 inHG
Pitot Leak Check, I: Pass
Pitot Leak Check, F: Pass
Probe ID: Glass M5 11ft B
Pitot ID: A8482
Filter ID Number: 5B447
Project Number: 25-012

Point	Sample Time	DGM (ft ³)	Delta H	Vacuum (in. Hg)	Delta P	Stack Temp. °F		Probe Temp °F		Filter Temp. °F		CPM Filter Temp. °F		Cord Box Temp	DGM Temp. °F
A-1	12:24	157.728	1.90	5	0.57	124		321		320		NA		61	66
2	12:28	160.780	1.90	5	0.56	124		321		320		NA		49	66
3	12:32	163.760	1.60	5	0.49	124		318		320		NA		48	67
B-1	12:40	166.525	1.90	5	0.58	124		320		320		NA		49	67
2	12:44	169.520	1.90	5	0.57	124		321		320		NA		50	67
3	12:48	172.510	1.60	5	0.49	124		318		320		NA		53	68
C-1	12:56	175.290	2.00	5	0.59	124		322		320		NA		55	69
2	13:00	178.340	1.90	5	0.57	124		319		320		NA		59	70
3	13:04	181.400	1.60	5	0.48	124		321		320		NA		62	72
D-1	13:11	184.215	2.00	5	0.58	124		320		319		NA		60	72
2	13:15	187.370	1.90	5	0.57	123		318		320		NA		58	72
3	13:19	190.310	1.60	5	0.48	123		321		320		NA		58	73
End	13:23	193.102													
2974.3		Total ft ³ :	Avg:	High:	Avg Sq. Rt.:	High:	Low:	High:	Low:	High:	Low:	High:	Low:	High:	Avg:
		35.374	1.82	5	0.7371	124	123	322	318	320	319			62	69

Initial Wt: 2974.3
Final Wt: 3091.6
Weight Gain: 117.3

DGM Calibration Check (Yqa): 1.0
(Yqa must be ± 5 to be valid)

K-Factor: 3.34
% ISO: 99.1

Evergy - Field Data

Plant, Unit: La Cygne Unit 1
Run #: 12
Date: 2/21/2025
Operator: SB
Test Type: RCA M5
Run Time: 48 minutes
Start/End: 14:04 15:04
Estimated Final
O₂, %: 9.00 7.90
CO₂, %: 10.50 11.63

Barometric Pressure: 29.12 inHG
Static Pressure: -0.75 inH₂O
Stack Diameter/Area: 375 in. / 766.99 sqft
itot Tube Coefficient: 0.84
DGM Coefficient (Y): 0.986
Delta H @: 1.949
Meter Box #: 1806015
Nozzle Diameter: 0.256
Assumed % Moisture: 16
Final % Moisture: 13.5 Saturated

Leak Check, I: 0.000 @ 15 inHG
Leak Check, F: 0.000 @ 5 inHG
Pitot Leak Check, I: Pass
Pitot Leak Check, F: Pass
Probe ID: Glass M5 11ft A
Pitot ID: A8481
Filter ID Number: 5B448
Project Number: 25-012

Point	Sample Time	DGM (ft³)	Delta H	Vacuum (in. Hg)	Delta P	Stack Temp. °F	Probe Temp °F	Filter Temp. °F	CPM Filter Temp. °F	Cold Box Temp	DGM Temp. °F		
A-1	14:04	194.278	2.00	5	0.60	124	322	320	NA	55	61		
2	14:08	197.360	1.90	5	0.57	124	319	319	NA	45	67		
3	14:12	200.320	1.90	5	0.55	124	321	320	NA	44	67		
B-1	14:20	203.330	2.20	5	0.66	124	320	320	NA	45	68		
2	14:24	206.510	2.20	5	0.64	124	320	320	NA	47	69		
3	14:28	209.710	2.20	5	0.63	124	320	320	NA	47	70		
C-1	14:36	212.920	2.50	5	0.74	124	320	320	NA	49	71		
2	14:40	216.340	2.50	5	0.72	125	318	320	NA	50	71		
3	14:44	219.790	2.40	5	0.70	125	319	320	NA	51	72		
D-1	14:52	223.160	2.80	5	0.82	125	321	320	NA	52	73		
2	14:56	226.860	2.70	5	0.80	125	320	320	NA	53	74		
3	15:00	230.470	2.70	5	0.78	125	319	320	NA	54	75		
End	15:04	234.090											
2982.2		Total ft³:	Avg:	High:	Avg Sq. Rt.:	High:	Low:	High:	Low:	High:	Low:	High:	Avg:
		39.812	2.15	5	0.8255	125	124	322	318	320	319		

Initial Wt: 2982.2
Final Wt: 3120.9
Weight Gain: 138.7

DGM Calibration Check (Yqa): 4.3
(Yqa must be ± 5 to be valid)

K-Factor: 3.39
% ISO: 99.4

Evergy - Field Data

Plant, Unit: La Cygne Unit 1
Run #: 13
Date: 2/21/2025
Operator: SB
Test Type: RCA M5
Run Time: 48 minutes
Start/End: 15:30 16:40
Estimated Final
O₂, %: 6.00 6.98
CO₂, %: 12.50 12.56

Barometric Pressure: 29.19 inHG
Static Pressure: -0.78 inH₂O
Stack Diameter/Area: 375 in. / 766.99 sqft
itot Tube Coefficient: 0.84
DGM Coefficient (Y): 0.986
Delta H @: 1.949
Meter Box #: 1806015
Nozzle Diameter: 0.226
Assumed % Moisture: 16
Final % Moisture: 15 Saturated

Leak Check, I: 0.000 @ 15 inHG
Leak Check, F: 0.000 @ 6 inHG
Pitot Leak Check, I: Pass
Pitot Leak Check, F: Pass
Probe ID: Glass M5 11ft B
Pitot ID: A8482
Filter ID Number: 5B449
Project Number: 25-012

Point	Sample Time	DGM (ft ³)	Delta H	Vacuum (in. Hg)	Delta P	Stack Temp. °F	Probe Temp °F	Filter Temp. °F	CPM Filter Temp. °F	Cord Box Temp	DGM Temp. °F			
A-1	15:30	235.575	2.00	5	0.97	128	320	320	NA	55	70			
2	15:34	238.590	2.10	5	1.00	128	320	320	NA	48	70			
3	15:38	241.720	2.00	5	0.95	129	321	320	NA	48	70			
B-1	15:46	244.790	2.30	5	1.10	129	319	321	NA	52	71			
2	15:50	247.990	2.00	5	0.97	129	319	320	NA	55	72			
3	15:54	251.105	2.00	5	0.95	129	320	320	NA	59	73			
C-1	16:02	254.115	2.00	5	0.98	129	320	320	NA	63	73			
2	16:06	257.245	2.00	6	0.97	129	320	320	NA	65	75			
3	16:20	260.310	1.90	6	0.94	129	318	321	NA	64	72			
D-1	16:28	263.300	2.10	6	1.00	129	320	321	NA	48	73			
2	16:32	266.395	2.10	6	1.00	128	318	320	NA	49	73			
3	16:36	269.560	2.00	6	0.96	129	319	320	NA	48	73			
End	16:40	271.992												
2947.6		Total ft ³ :	Avg:	High:	Avg Sq. Rt.:		High:	Low:	High:	Low:	High:	Low:	High:	Avg:
		36.417	2.04	6	0.9910	129	128	321	318	321	320		65	72

Initial Wt: 2947.6
Final Wt: 3092.6
Weight Gain: 145

DGM Calibration Check (Yqa): -1.8
(Yqa must be ± 5 to be valid)

K-Factor: 2.06
% ISO: 99.1

Evergy - Field Data

Plant, Unit: La Cygne Unit 1
 Run #: 14
 Date: 2/21/2025
 Operator: SB
 Test Type: RCA M5
 Run Time: 48 minutes
 Start/End: 17:22 18:18
 Estimated Final
 O₂, %: 6.00 7.11
 CO₂, %: 14.00 12.54

Barometric Pressure: 29.19 inHG
 Static Pressure: -0.78 inH₂O
 Stack Diameter/Area: 375 in. / 766.99 sqft
 itot Tube Coefficient: 0.84
 DGM Coefficient (Y): 0.986
 Delta H @: 1.949
 Meter Box #: 1806015
 Nozzle Diameter: 0.226
 Assumed % Moisture: 16
 Final % Moisture: 15.2 Saturated

Leak Check, I: 0.000 @ 15 inHG
 Leak Check, F: 0.000 @ 5 inHG
 Pitot Leak Check, I: Pass
 Pitot Leak Check, F: Pass
 Probe ID: Glass M5 11ft A
 Pitot ID: A8481
 Filter ID Number: 5B450
 Project Number: 25-012

Point	Sample Time	DGM (ft³)	Delta H	Vacuum (in. Hg)	Delta P	Stack Temp. °F	Probe Temp °F	Filter Temp. °F	CPM Filter Temp. °F	Cord Box Temp.	DGM Temp. °F		
A-1	17:22	272.720	2.30	5	1.10	129	321	320	NA	49	71		
2	17:26	276.050	2.10	5	1.00	129	319	321	NA	52	72		
3	17:30	279.130	2.00	5	0.99	129	320	321	NA	50	72		
B-1	17:38	282.240	2.50	5	1.20	129	319	320	NA	49	73		
2	17:42	285.570	2.30	5	1.10	129	320	321	NA	49	74		
3	17:46	288.840	2.00	5	0.97	129	320	320	320	50	75		
C-1	17:54	291.930	2.50	5	1.20	129	320	321	NA	52	75		
2	17:58	295.320	2.30	5	1.10	129	321	321	NA	53	75		
3	18:02	298.630	1.90	5	0.92	129	320	320	NA	55	77		
D-1	18:06	301.660	2.50	5	1.20	129	320	320	NA	58	77		
2	18:10	305.030	2.10	5	1.00	129	320	321	NA	60	78		
3	18:14	308.170	2.00	5	0.97	129	320	320	NA	62	79		
End	18:18	311.218											
2973.5		Total ft³:	Avg:	High:	Avg Sq. Rt.:	High:	Low:	High:	Low:	High:	Low:	High:	Avg:
		38.498	2.21	5	1.0297	129	129	321	319	321	320		62

Initial Wt: 2973.5
 Final Wt: 3124.1
 Weight Gain: 150.6

DGM Calibration Check (Yqa): -0.4
 (Yqa must be ± 5 to be valid)

K-Factor: 2.07
 % ISO: 100.2

Evergy - Field Data

Plant/Unit: La Cygne Unit 1 Project Number: 25-012
 Operator: SB Date: 2/19/2025

Nozzle Set	Nozzle ID	D _n (in)		
G-1	7	0.226		
G-2	7	0.221		
Pitot ID	Evidence of damage?	Evidence of mis-alignment?	Calibration or Repair required?	
A8481	No	No	No	
A8482	No	No	No	
Probe ID	Reference Temp. (°F)	Indicated Temp. (°F)	Difference	Criteria
Glass M5 11 ft A	66.0	65.0	0.2%	± 1.5 % (absolute)
Glass M5 11 ft B	66.0	66.0	0.0%	
Barometric Reading from:	Barometric Reading @ MSL (inHG)	Barometric Reading @ Elevation	Elevation (ft)	
NOAA	29.67	29.23	440	

Plant/Unit: La Cygne Unit 1
 Project No.: 25-012

Method 4 Moisture Determination

Impinger Weights

		Contents	Initial Weight	Final Weight	Difference
Run 1	#1	DI Water	705.9	804	98.1
	#2	DI Water	698.6	738.2	39.6
	#3	Empty	609.6	616.8	7.2
	#4	Silica	974.1	993.4	19.3
		Total:	2988.2	3152.4	
Run 2	#1	DI Water	708.4	866.1	157.7
	#2	DI Water	694.9	658.8	-36.1
	#3	Empty	616.8	619.4	2.6
	#4	Silica	993.4	1002.5	9.1
		Total:	3013.5	3146.8	
Run 3	#1	DI Water	730.6	837.4	106.8
	#2	DI Water	658.8	679.7	20.9
	#3	Empty	619.4	622.9	3.5
	#4	Silica	1002.5	1012.2	9.7
		Total:	3011.3	3152.2	
Run 4	#1	DI Water	707.5	811.1	103.6
	#2	DI Water	679.7	706	26.3
	#3	Empty	622.9	625.5	2.6
	#4	Silica	1012.2	1020.4	8.2
		Total:	3022.3	3163	
Run 5	#1	DI Water	682.9	789.2	106.3
	#2	DI Water	671.2	696.1	24.9
	#3	Empty	625.5	628.5	3
	#4	Silica	1020.4	1028.7	8.3
		Total:	3000	3142.5	
Run 6	#1	DI Water	720	827	107
	#2	DI Water	696.1	721.8	25.7
	#3	Empty	628.5	631.8	3.3
	#4	Silica	1028.7	1035.8	7.1
		Total:	3073.3	3216.4	

Run 7	#1	Empty	424	683	259
	#2	Empty	601.9	622.7	20.8
	#3	Empty	599.8	625.2	25.4
	#4	Silica	924.9	958.8	33.9
	Total:		2550.6	2889.7	
Run 8	#1	Empty	417.8	660.5	242.7
	#2	Empty	601.5	627.6	26.1
	#3	Empty	612.6	645.6	33
	#4	Silica	939.2	976.6	37.4
	Total:		2571.1	2910.3	
Run 9	#1	Empty	425.3	699.1	273.8
	#2	Empty	603.6	620.1	16.5
	#3	Empty	602.6	628.5	25.9
	#4	Silica	958.2	992.2	34
	Total:		2589.7	2939.9	
Run 10	#1	DI Water	720.1	804.9	84.8
	#2	DI Water	667.7	691.6	23.9
	#3	Empty	613.6	617.6	4
	#4	Silica	945.6	959	13.4
	Total:		2947	3073.1	
Run 11	#1	DI Water	710.9	792.5	81.6
	#2	DI Water	691.6	713.6	22
	#3	Empty	612.8	615.9	3.1
	#4	Silica	959	969.6	10.6
	Total:		2974.3	3091.6	
Run 12	#1	DI Water	686.1	779.5	93.4
	#2	DI Water	713.6	743.5	29.9
	#3	Empty	612.9	617.2	4.3
	#4	Silica	969.6	980.7	11.1
	Total:		2982.2	3120.9	
Run 13	#1	DI Water	691.8	795.5	103.7
	#2	DI Water	657.9	688.1	30.2
	#3	Empty	617.2	620.7	3.5
	#4	Silica	980.7	988.3	7.6
	Total:		2947.6	3092.6	
Run 14	#1	DI Water	705	812.1	107.1
	#2	DI Water	688.1	713.1	25
	#3	Empty	615.9	619.3	3.4
	#4	Silica	964.5	979.6	15.1
	Total:		2973.5	3124.1	

INSTRUMENT VALIDATION CHECK

T802 Final Calibrated Test and Validation Data

5

12/11/18

Model:	T802	Serial Number:	155	Sales Order:	
Firmware:	063020001 1.0.2.73	Technician:	FG	SP#:	YES
Date:	12/10/2018				

Parameter	Displayed As	Observed Value	Units	Final Test Process Control Limits at Factory**
O ₂ Range	O ₂ RNG	0-25	WT%	0 - 1 WT% to 1 - 100 WT%
CO ₂ Range (Opt only)	CO ₂ RNG	0-20	WT%	0 - 20 WT% Standard
Stability	O ₂ STB	0	WT%	< 0.05 % with Zero Air
Sample Pressure	PRES	29.2	In-Hg-A	Ambient Pressure ± 1.0 In Hg
Sample Flow	SAMP FL	138	cc/min	120 ± 20
O ₂ Slope	O ₂ SLOPE	1.009	-	1.0 ± 0.1
O ₂ Offset	O ₂ OFST	0.27	%	≤ 1%
CO ₂ Slope (Opt Only)	CO ₂ SLOPE	1.015	-	1.0 ± 0.1
CO ₂ Offset (Opt Only)	CO ₂ OFST	0.08	%	≤ 1%
O ₂ Cell Temperature *	O ₂ CELL TEMP	50	°C	50 ± .5
CO ₂ Cell Temp * (Opt Only)	CO ₂ CELL TEMP	50	°C	50 ± .5
Box Temperature	BOX TEMP	32.9	°C	31 ± 4
Time of Day	TIME	13:59:57	HH:MM:SS	

* For good instrument performance, the steadiness of this signal is more important than its absolute value (within the operating range)

** these are process control limits, and not specification limits. Items out of range do not imply the unit is out of specification.

Statement of Calibration

The unit identified above has been tested with NIST measuring and test equipment using lot traceable materials. The testing is performed in accordance with ISO 9001:2015 and is traceable to NIST and industry recognized standards.

Configuration and Options

Power Configuration: Voltage	100-125
Frequency	60 Hz
HV internal pump (13)	
Rack Mounts with Slides (20A - 26"; 20B - 24")	
Rack Mounts Ears Only (21)	
Strap Carrying Handle (29)	
Multi-drop RS-232 (62)	
Ethernet (63A)	Yes
Multi-drop/ethernet combo (63C)	
USB Com 2 T-series only (64A)	
Analog Input T-series only (64B)	
COM 2 Configuration	
Warranty Extended 1 Year (92A)	
A1 Voltage Output	
A2 Voltage Output	
A3 Voltage Output	
A4 Voltage Output	5
Current Output, Channel A1 (41)	4 - 20 mA
Current Output, Channel A2 (41)	4 - 20 mA
Current Output, Channel A3 (41)	

Additional Information:

TML, OPT, SLIDES WITH EXT, PN: 023600100. NUMAVIEW FW, PN: 062910000 1.4.4.115. OPT 67A, CO2 SENSOR PN: 064790000

FACILITY CEMS DATA

Date/Time	BOILER01 LOAD (MW) Value	BOILER01 PM_RAW (mg/scm) Value	Date/Time	BOILER01 LOAD (MW) Value	BOILER01 PM_RAW (mg/scm) Value
02/19/2025 10:34	502	2.4	02/19/2025 12:00	763	3.2
02/19/2025 10:35	501	2.6	02/19/2025 12:01	763	2.9
02/19/2025 10:36	501	2.5	02/19/2025 12:02	764	3
02/19/2025 10:37	500	2.4	02/19/2025 12:03	769	2.9
02/19/2025 10:38	500	2.8	02/19/2025 12:04	778	2.8
02/19/2025 10:39	500	2.6	02/19/2025 12:05	788	3
02/19/2025 10:40	500	2.6	02/19/2025 12:06	792	2.7
02/19/2025 10:41	501	2.6	02/19/2025 12:07	794	2.9
02/19/2025 10:42	503	2.6	02/19/2025 12:08	800	3.1
02/19/2025 10:43	506	2.7	02/19/2025 12:09	805	3
02/19/2025 10:44	509	2.5	02/19/2025 12:10	811	3
02/19/2025 10:45	512	2.7	02/19/2025 12:11	817	2.9
02/19/2025 10:46	516	2.6	02/19/2025 12:12	819	2.9
02/19/2025 10:47	518	2.5	02/19/2025 12:13	815	2.7
02/19/2025 10:48	522	2.9	02/19/2025 12:14	815	2.8
02/19/2025 10:49	528	2.6	02/19/2025 12:15	814	2.8
02/19/2025 10:50	533	2.9	02/19/2025 12:16	819	2.6
02/19/2025 10:51	542	2.8	02/19/2025 12:17	822	3
02/19/2025 10:52	551	2.8	02/19/2025 12:18	822	2.7
02/19/2025 10:53	554	2.9	02/19/2025 12:19	823	2.7
02/19/2025 10:54	558	2.7	02/19/2025 12:20	824	2.8
02/19/2025 10:55	560	3	02/19/2025 12:21	823	2.6
02/19/2025 10:56	562	3.1	02/19/2025 12:22	821	2.8
02/19/2025 10:57	566	3.2	02/19/2025 12:23	823	2.6
02/19/2025 10:58	575	3.2	02/19/2025 12:24	823	2.8
02/19/2025 10:59	584	3.3	02/19/2025 12:25	821	2.9
02/19/2025 11:00	591	3.3	02/19/2025 12:26	819	3
02/19/2025 11:01	596	3.2	02/19/2025 12:27	819	3.1
02/19/2025 11:02	601	3.2	02/19/2025 12:28	818	3.1
02/19/2025 11:03	603	3.2	02/19/2025 12:29	817	3.1
02/19/2025 11:04	609	3.2	02/19/2025 12:30	817	3.1
02/19/2025 11:05	619	3.2	02/19/2025 12:31	818	3.1
02/19/2025 11:06	627	3.2	02/19/2025 12:32	818	3.3
02/19/2025 11:07	632	3.4	02/19/2025 12:33	819	3.2
02/19/2025 11:08	635	3.4	02/19/2025 12:34	819	3.2
02/19/2025 11:09	638	3.4	02/19/2025 12:35	818	3.3
02/19/2025 11:10	645	3.6	02/19/2025 12:36	819	3.3
02/19/2025 11:11	650	3.5	02/19/2025 12:37	819	3.5
02/19/2025 11:12	652	3.5	02/19/2025 12:38	820	3.3
02/19/2025 11:13	655	3.3	02/19/2025 12:39	820	3.3
02/19/2025 11:14	660	3.3	02/19/2025 12:40	819	3.2
02/19/2025 11:15	666	3.3	02/19/2025 12:41	820	3.4
02/19/2025 11:16	671	3.1	02/19/2025 12:42	819	3.5
02/19/2025 11:17	676	3.1	02/19/2025 12:43	819	3.5
02/19/2025 11:18	679	3.3	02/19/2025 12:44	820	3.6
02/19/2025 11:19	682	3	02/19/2025 12:45	821	3.5
02/19/2025 11:20	686	3.2	02/19/2025 12:46	820	3.3
02/19/2025 11:21	691	2.9	02/19/2025 12:47	820	3.4
02/19/2025 11:22	697	3	02/19/2025 12:48	820	3.1
02/19/2025 11:23	701	3.2	02/19/2025 12:49	820	3.1
02/19/2025 11:24	705	3.1	02/19/2025 12:50	821	3.1
02/19/2025 11:25	709	3.2	02/19/2025 12:51	821	2.9
02/19/2025 11:26	714	2.8	02/19/2025 12:52	820	3.1
02/19/2025 11:27	719	2.9	02/19/2025 12:53	819	3
02/19/2025 11:28	727	2.9	02/19/2025 12:54	819	3.1
02/19/2025 11:29	729	2.6	02/19/2025 12:55	819	3.1
02/19/2025 11:30	737	3	02/19/2025 12:56	819	2.6
02/19/2025 11:31	743	2.5	02/19/2025 12:57	819	2.7
02/19/2025 11:32	748	2.7	02/19/2025 12:58	819	2.4
02/19/2025 11:33	751	2.8	02/19/2025 12:59	818	2.4
02/19/2025 11:34	752	2.5	02/19/2025 13:00	817	2.4
02/19/2025 11:35	754	2.8	02/19/2025 13:01	816	2.2
02/19/2025 11:36	756	2.6	02/19/2025 13:02	814	2.4
02/19/2025 11:37	758	2.7	02/19/2025 13:03	815	2.3
Grand Summaries			Grand Summaries		
	Avg: 618	Avg: 2.9		Avg: 813	Avg: 3.0
	Count: 64	Count: 64		Count: 64	Count: 64

Date/Time	BOILER01 LOAD (MW) Value	BOILER01 PM_RAW (mg/scm) Value	Date/Time	BOILER01 LOAD (MW) Value	BOILER01 PM_RAW (mg/scm) Value
02/19/2025 13:12	818	2.4	02/19/2025 14:48	820	3.1
02/19/2025 13:13	818	2.4	02/19/2025 14:49	820	3.1
02/19/2025 13:14	817	2.5	02/19/2025 14:50	819	3.3
02/19/2025 13:15	817	2.6	02/19/2025 14:51	818	3.5
02/19/2025 13:16	817	2.6	02/19/2025 14:52	818	3.2
02/19/2025 13:17	817	2.6	02/19/2025 14:53	820	3.4
02/19/2025 13:18	817	2.6	02/19/2025 14:54	819	3.4
02/19/2025 13:19	816	2.6	02/19/2025 14:55	818	3.1
02/19/2025 13:20	817	2.6	02/19/2025 14:56	817	3.2
02/19/2025 13:21	817	2.7	02/19/2025 14:57	818	3.2
02/19/2025 13:22	816	2.9	02/19/2025 14:58	822	3.3
02/19/2025 13:23	817	2.8	02/19/2025 14:59	820	3.2
02/19/2025 13:24	817	2.9	02/19/2025 15:00	819	3
02/19/2025 13:25	817	2.9	02/19/2025 15:01	819	3.1
02/19/2025 13:26	817	2.9	02/19/2025 15:02	819	3
02/19/2025 13:27	816	2.8	02/19/2025 15:03	819	2.9
02/19/2025 13:28	817	2.9	02/19/2025 15:04	820	2.9
02/19/2025 13:29	818	2.9	02/19/2025 15:05	821	2.7
02/19/2025 13:30	817	2.8	02/19/2025 15:06	821	2.9
02/19/2025 13:31	816	2.7	02/19/2025 15:07	820	2.9
02/19/2025 13:32	814	2.8	02/19/2025 15:08	821	3.1
02/19/2025 13:33	812	2.6	02/19/2025 15:09	821	3.2
02/19/2025 13:34	811	2.8	02/19/2025 15:10	821	3
02/19/2025 13:35	812	2.7	02/19/2025 15:11	819	3.1
02/19/2025 13:36	814	2.6	02/19/2025 15:12	821	2.8
02/19/2025 13:37	818	2.9	02/19/2025 15:13	820	2.9
02/19/2025 13:38	815	2.8	02/19/2025 15:14	821	2.9
02/19/2025 13:39	814	2.9	02/19/2025 15:15	820	2.7
02/19/2025 13:40	817	2.7	02/19/2025 15:16	820	3
02/19/2025 13:41	817	2.6	02/19/2025 15:17	821	2.7
02/19/2025 13:42	818	2.7	02/19/2025 15:18	821	2.8
02/19/2025 13:43	819	2.4	02/19/2025 15:19	821	2.9
02/19/2025 13:44	817	2.7	02/19/2025 15:20	820	2.8
02/19/2025 13:45	817	2.6	02/19/2025 15:21	820	3
02/19/2025 13:46	818	2.5	02/19/2025 15:22	821	2.8
02/19/2025 13:47	818	2.6	02/19/2025 15:23	820	2.9
02/19/2025 13:48	819	2.4	02/19/2025 15:24	819	3.1
02/19/2025 13:49	819	2.6	02/19/2025 15:25	821	3.1
02/19/2025 13:50	816	2.4	02/19/2025 15:26	826	3.2
02/19/2025 13:51	816	2.5	02/19/2025 15:27	818	3.2
02/19/2025 13:52	819	2.5	02/19/2025 15:28	813	3.3
02/19/2025 13:53	818	2.4	02/19/2025 15:29	817	3.3
02/19/2025 13:54	818	2.7	02/19/2025 15:30	820	3.4
02/19/2025 13:55	819	2.8	02/19/2025 15:31	820	3.3
02/19/2025 13:56	817	2.9	02/19/2025 15:32	819	3.2
02/19/2025 13:57	817	2.9	02/19/2025 15:33	820	3.1
02/19/2025 13:58	818	3	02/19/2025 15:34	820	3.2
02/19/2025 13:59	817	3	02/19/2025 15:35	819	3.2
02/19/2025 14:00	817	3	02/19/2025 15:36	820	3.1
02/19/2025 14:01	818	2.9	02/19/2025 15:37	821	3.1
02/19/2025 14:02	816	3	02/19/2025 15:38	822	3
02/19/2025 14:03	817	3.1	02/19/2025 15:39	821	3.2
02/19/2025 14:04	814	3.1	02/19/2025 15:40	820	3.1
02/19/2025 14:05	813	3.1	02/19/2025 15:41	820	3.2
02/19/2025 14:06	816	3.1	02/19/2025 15:42	820	3.4
02/19/2025 14:07	820	3.1	02/19/2025 15:43	821	3.4
02/19/2025 14:08	813	3.2	02/19/2025 15:44	821	3.3
02/19/2025 14:09	812	3.2	02/19/2025 15:45	821	3.1
02/19/2025 14:10	817	3.2	02/19/2025 15:46	824	3.3
02/19/2025 14:11	816	3	02/19/2025 15:47	819	3
02/19/2025 14:12	816	3.1	02/19/2025 15:48	820	3.1
02/19/2025 14:13	819	3.2	02/19/2025 15:49	821	3.1
02/19/2025 14:14	814	3.1	02/19/2025 15:50	822	2.9
02/19/2025 14:15	814	2.9	02/19/2025 15:51	823	3.1
Grand Summaries			Grand Summaries		
	Avg: 816	Avg: 2.8		Avg: 820	Avg: 3.1
	Count: 64	Count: 64		Count: 64	Count: 64

Date/Time	BOILER01 LOAD (MW) Value	BOILER01 PM_RAW (mg/scm) Value	Date/Time	BOILER01 LOAD (MW) Value	BOILER01 PM_RAW (mg/scm) Value
02/19/2025 16:00	819	2.8	02/19/2025 17:07	822	3.3
02/19/2025 16:01	819	3	02/19/2025 17:08	822	3
02/19/2025 16:02	820	2.7	02/19/2025 17:09	822	3.3
02/19/2025 16:03	820	2.9	02/19/2025 17:10	823	3.3
02/19/2025 16:04	820	2.8	02/19/2025 17:11	822	3.3
02/19/2025 16:05	822	2.8	02/19/2025 17:12	822	3.3
02/19/2025 16:06	822	3.2	02/19/2025 17:13	823	3.3
02/19/2025 16:07	820	2.7	02/19/2025 17:14	822	3
02/19/2025 16:08	820	3	02/19/2025 17:15	822	3.1
02/19/2025 16:09	821	3.1	02/19/2025 17:16	822	3
02/19/2025 16:10	821	3.2	02/19/2025 17:17	822	3
02/19/2025 16:11	821	3.2	02/19/2025 17:18	822	3.2
02/19/2025 16:12	821	3.2	02/19/2025 17:19	822	2.8
02/19/2025 16:13	821	3.2	02/19/2025 17:20	821	3.1
02/19/2025 16:14	820	3.3	02/19/2025 17:21	820	3.1
02/19/2025 16:15	819	3.3	02/19/2025 17:22	820	3.1
02/19/2025 16:16	818	3.3	02/19/2025 17:23	820	3.2
02/19/2025 16:17	818	3.4	02/19/2025 17:24	820	3
02/19/2025 16:18	819	3.4	02/19/2025 17:25	821	3.1
02/19/2025 16:19	816	3.3	02/19/2025 17:26	821	2.9
02/19/2025 16:20	818	3.3	02/19/2025 17:27	821	2.8
02/19/2025 16:21	820	3.1	02/19/2025 17:28	822	3
02/19/2025 16:22	821	3	02/19/2025 17:29	821	2.7
02/19/2025 16:23	819	3.2	02/19/2025 17:30	820	3
02/19/2025 16:24	820	3.4	02/19/2025 17:31	817	2.8
02/19/2025 16:25	820	3.4	02/19/2025 17:32	818	2.8
02/19/2025 16:26	820	3.5	02/19/2025 17:33	818	2.9
02/19/2025 16:27	821	3.5	02/19/2025 17:34	818	2.7
02/19/2025 16:28	821	3.5	02/19/2025 17:35	816	3
02/19/2025 16:29	821	3.2	02/19/2025 17:36	818	2.7
02/19/2025 16:30	820	3.2	02/19/2025 17:37	819	2.8
02/19/2025 16:31	821	3.2	02/19/2025 17:38	818	3.1
02/19/2025 16:32	821	2.8	02/19/2025 17:39	817	3.1
02/19/2025 16:33	821	3.2	02/19/2025 17:40	817	3.2
02/19/2025 16:34	822	2.8	02/19/2025 17:41	819	2.9
02/19/2025 16:35	822	2.7	02/19/2025 17:42	819	3.1
02/19/2025 16:36	823	2.9	02/19/2025 17:43	819	3.3
02/19/2025 16:37	822	2.8	02/19/2025 17:44	820	3.2
02/19/2025 16:38	821	2.9	02/19/2025 17:45	820	3
02/19/2025 16:39	823	2.7	02/19/2025 17:46	820	3
02/19/2025 16:40	823	3	02/19/2025 17:47	822	3.1
02/19/2025 16:41	819	2.9	02/19/2025 17:48	821	3
02/19/2025 16:42	820	2.7	02/19/2025 17:49	821	3
02/19/2025 16:43	823	3	02/19/2025 17:50	822	3
02/19/2025 16:44	821	2.7	02/19/2025 17:51	822	3
02/19/2025 16:45	820	2.8	02/19/2025 17:52	820	3.1
02/19/2025 16:46	820	2.9	02/19/2025 17:53	819	3.2
02/19/2025 16:47	820	2.6	02/19/2025 17:54	818	3.1
02/19/2025 16:48	821	2.6	02/19/2025 17:55	817	3.1
02/19/2025 16:49	819	2.5	02/19/2025 17:56	815	3.2
02/19/2025 16:50	818	2.9	02/19/2025 17:57	809	3.1
02/19/2025 16:51	818	2.8	02/19/2025 17:58	804	3
02/19/2025 16:52	819	2.7	02/19/2025 17:59	800	3
02/19/2025 16:53	817	3.1	02/19/2025 18:00	799	3.2
02/19/2025 16:54	816	3.1	02/19/2025 18:01	800	2.8
02/19/2025 16:55	816	3.1	02/19/2025 18:02	801	2.8
02/19/2025 16:56	816	3.2	02/19/2025 18:03	801	3
02/19/2025 16:57	816	3.2	02/19/2025 18:04	801	2.9
02/19/2025 16:58	817	3.2	02/19/2025 18:05	802	3.1
02/19/2025 16:59	817	3.3	02/19/2025 18:06	805	2.9
02/19/2025 17:00	818	3.2	02/19/2025 18:07	810	2.9
02/19/2025 17:01	820	3.2	02/19/2025 18:08	814	3.1
02/19/2025 17:02	821	3.3	02/19/2025 18:09	817	3
02/19/2025 17:03	820	3.3	02/19/2025 18:10	817	3.1
Grand Summaries			Grand Summaries		
	Avg: 820	Avg: 3.1		Avg: 817	Avg: 3.0
	Count: 64	Count: 64		Count: 64	Count: 64

Date/Time	BOILER01 LOAD (MW) Value	BOILER01 PM_RAW (mg/scm) Value			
02/20/2025 13:00	817	2.8	02/20/2025 14:03	820	3
02/20/2025 13:01	817	2.9	02/20/2025 14:04	819	3.1
02/20/2025 13:02	817	3	02/20/2025 14:05	818	3.1
02/20/2025 13:03	818	2.8	02/20/2025 14:06	819	3.2
02/20/2025 13:04	818	3	02/20/2025 14:07	819	3.2
02/20/2025 13:05	818	2.6	02/20/2025 14:08	819	3.2
02/20/2025 13:06	818	2.7	02/20/2025 14:09	819	3.3
02/20/2025 13:07	819	2.8	02/20/2025 14:10	819	3.4
02/20/2025 13:08	820	2.6	02/20/2025 14:11	819	3.3
02/20/2025 13:09	820	2.9	02/20/2025 14:12	821	3.2
02/20/2025 13:10	820	2.6	02/20/2025 14:13	820	3
02/20/2025 13:11	819	2.8	02/20/2025 14:14	819	2.9
02/20/2025 13:12	820	2.7	02/20/2025 14:15	818	3
02/20/2025 13:13	820	2.5	02/20/2025 14:16	819	3.3
02/20/2025 13:14	820	2.8	02/20/2025 14:17	819	3.1
02/20/2025 13:15	820	2.3	02/20/2025 14:18	818	3.1
02/20/2025 13:16	820	2.4	02/20/2025 14:19	817	3.1
02/20/2025 13:17	820	2.6	02/20/2025 14:20	817	2.8
02/20/2025 13:18	819	2.6	02/20/2025 14:21	818	2.9
02/20/2025 13:19	819	2.8	02/20/2025 14:22	818	2.7
02/20/2025 13:20	819	3	02/20/2025 14:23	817	2.8
02/20/2025 13:21	819	3.1	02/20/2025 14:24	816	3.1
02/20/2025 13:22	819	3	02/20/2025 14:25	817	3.2
02/20/2025 13:23	819	3	02/20/2025 14:26	817	3.2
02/20/2025 13:24	819	3.1	02/20/2025 14:27	818	3.1
02/20/2025 13:25	820	3	02/20/2025 14:28	818	3
02/20/2025 13:26	819	2.9	02/20/2025 14:29	818	3
02/20/2025 13:27	819	2.9	02/20/2025 14:30	819	2.6
02/20/2025 13:28	819	2.8	02/20/2025 14:31	820	2.9
02/20/2025 13:29	819	2.9	02/20/2025 14:32	820	2.7
02/20/2025 13:30	819	3	02/20/2025 14:33	821	2.8
02/20/2025 13:31	819	2.9	02/20/2025 14:34	822	2.9
02/20/2025 13:32	819	2.8	02/20/2025 14:35	822	2.7
02/20/2025 13:33	820	3	02/20/2025 14:36	822	2.9
02/20/2025 13:34	820	3	02/20/2025 14:37	822	2.7
02/20/2025 13:35	821	3	02/20/2025 14:38	823	2.8
02/20/2025 13:36	820	3	02/20/2025 14:39	822	2.8
02/20/2025 13:37	819	2.8	02/20/2025 14:40	822	2.7
02/20/2025 13:38	819	2.7	02/20/2025 14:41	823	2.9
02/20/2025 13:39	820	2.9	02/20/2025 14:42	823	3
02/20/2025 13:40	820	2.7	02/20/2025 14:43	823	3.1
02/20/2025 13:41	820	2.9	02/20/2025 14:44	823	3.1
02/20/2025 13:42	819	3	02/20/2025 14:45	822	3.3
02/20/2025 13:43	820	2.9	02/20/2025 14:46	821	3.2
02/20/2025 13:44	822	2.9	02/20/2025 14:47	821	3.2
02/20/2025 13:45	821	2.6	02/20/2025 14:48	822	3.2
02/20/2025 13:46	820	2.7	02/20/2025 14:49	822	3.2
02/20/2025 13:47	820	2.6	02/20/2025 14:50	822	3.3
02/20/2025 13:48	820	2.4	02/20/2025 14:51	821	3.2
02/20/2025 13:49	820	2.6	02/20/2025 14:52	823	3.2
02/20/2025 13:50	820	2.4	02/20/2025 14:53	822	3.3
02/20/2025 13:51	819	2.5	02/20/2025 14:54	823	3.3
02/20/2025 13:52	819	2.4	02/20/2025 14:55	822	3.3
02/20/2025 13:53	820	2.5	02/20/2025 14:56	822	3.3
02/20/2025 13:54	820	2.8	02/20/2025 14:57	822	3.3
02/20/2025 13:55	819	2.5	02/20/2025 14:58	822	3.4
02/20/2025 13:56	819	2.6	02/20/2025 14:59	820	3.4
02/20/2025 13:57	819	2.4	02/20/2025 15:00	819	3.4
02/20/2025 13:58	819	2.4	02/20/2025 15:01	818	3.4
02/20/2025 13:59	819	2.8	02/20/2025 15:02	818	3.2
02/20/2025 14:00	819	2.9	02/20/2025 15:03	818	3.1
02/20/2025 14:01	818	3.1	02/20/2025 15:04	818	3.1
02/20/2025 14:02	820	3.1	02/20/2025 15:05	818	2.9
			02/20/2025 15:06	818	3.2
			02/20/2025 15:07	819	3.2
			02/20/2025 15:08	820	3.2
			02/20/2025 15:09	820	3.1
			02/20/2025 15:10	819	3
			02/20/2025 15:11	819	3.1
			02/20/2025 15:12	820	2.7
			02/20/2025 15:13	820	2.8
			02/20/2025 15:14	821	2.8
			02/20/2025 15:15	822	2.7
			Grand Summaries		
				Avg: 820	Avg: 2.9
				Count: 136	Count: 136

Date/Time	BOILER01 LOAD (MW) Value	BOILER01 PM_RAW (mg/scm) Value			
02/20/2025 15:26	823	3.1	02/20/2025 16:29	817	3
02/20/2025 15:27	823	3.2	02/20/2025 16:30	817	3
02/20/2025 15:28	822	3.2	02/20/2025 16:31	818	3.2
02/20/2025 15:29	823	3.1	02/20/2025 16:32	819	3.1
02/20/2025 15:30	821	3.2	02/20/2025 16:33	821	3.3
02/20/2025 15:31	821	3.3	02/20/2025 16:34	821	3
02/20/2025 15:32	821	3.3	02/20/2025 16:35	822	3.1
02/20/2025 15:33	821	3.2	02/20/2025 16:36	822	3
02/20/2025 15:34	821	3.3	02/20/2025 16:37	822	2.7
02/20/2025 15:35	820	3.3	02/20/2025 16:38	823	2.9
02/20/2025 15:36	819	3.3	02/20/2025 16:39	822	2.7
02/20/2025 15:37	818	3.3	02/20/2025 16:40	822	2.9
02/20/2025 15:38	818	3.4	02/20/2025 16:41	821	2.9
02/20/2025 15:39	816	3.4	02/20/2025 16:42	822	2.7
02/20/2025 15:40	815	3.4	02/20/2025 16:43	822	3
02/20/2025 15:41	815	3.4	02/20/2025 16:44	821	2.7
02/20/2025 15:42	815	3.3	02/20/2025 16:45	822	2.9
02/20/2025 15:43	815	3.4	02/20/2025 16:46	821	2.8
02/20/2025 15:44	817	3.2	02/20/2025 16:47	820	2.7
02/20/2025 15:45	817	3.1	02/20/2025 16:48	819	3
02/20/2025 15:46	817	3.2	02/20/2025 16:49	819	3.2
02/20/2025 15:47	819	2.9	02/20/2025 16:50	818	3.2
02/20/2025 15:48	820	3.1	02/20/2025 16:51	819	3.2
02/20/2025 15:49	820	3.1	02/20/2025 16:52	818	3.2
02/20/2025 15:50	820	3.1	02/20/2025 16:53	818	3.2
02/20/2025 15:51	820	3.2	02/20/2025 16:54	819	3.3
02/20/2025 15:52	821	2.9	02/20/2025 16:55	819	3.3
02/20/2025 15:53	822	3.1	02/20/2025 16:56	819	3.3
02/20/2025 15:54	823	2.8	02/20/2025 16:57	820	3.3
02/20/2025 15:55	823	2.7	02/20/2025 16:58	820	3.3
02/20/2025 15:56	823	2.9	02/20/2025 16:59	820	3.4
02/20/2025 15:57	824	2.7	02/20/2025 17:00	820	3.4
02/20/2025 15:58	824	2.9	02/20/2025 17:01	820	3.3
02/20/2025 15:59	824	2.7	02/20/2025 17:02	821	3.4
02/20/2025 16:00	824	2.7	02/20/2025 17:03	821	3.4
02/20/2025 16:01	824	2.8	02/20/2025 17:04	821	3.5
02/20/2025 16:02	823	2.6	02/20/2025 17:05	822	3.4
02/20/2025 16:03	824	2.9	02/20/2025 17:06	822	3.4
02/20/2025 16:04	823	2.7	02/20/2025 17:07	821	3.5
02/20/2025 16:05	823	2.7	02/20/2025 17:08	823	3.4
02/20/2025 16:06	822	3	02/20/2025 17:09	823	3.1
02/20/2025 16:07	821	3.2	02/20/2025 17:10	822	3.2
02/20/2025 16:08	821	3.2	02/20/2025 17:11	822	3
02/20/2025 16:09	821	3.1	02/20/2025 17:12	822	3
02/20/2025 16:10	821	3.1	02/20/2025 17:13	822	3.1
02/20/2025 16:11	821	3.1	02/20/2025 17:14	822	3
02/20/2025 16:12	820	3.2	02/20/2025 17:15	822	3.1
02/20/2025 16:13	819	3.3	02/20/2025 17:16	822	3.1
02/20/2025 16:14	820	3.2	02/20/2025 17:17	822	3
02/20/2025 16:15	821	3.3	02/20/2025 17:18	822	3.1
02/20/2025 16:16	820	3.3	02/20/2025 17:19	822	2.7
02/20/2025 16:17	819	3.3	02/20/2025 17:20	822	2.9
02/20/2025 16:18	820	3.4	02/20/2025 17:21	822	2.8
02/20/2025 16:19	819	3.3	02/20/2025 17:22	824	2.8
02/20/2025 16:20	818	3.3	02/20/2025 17:23	823	2.9
02/20/2025 16:21	817	3.3	02/20/2025 17:24	823	2.7
02/20/2025 16:22	816	3.4	02/20/2025 17:25	823	2.9
02/20/2025 16:23	817	3.3	02/20/2025 17:26	823	2.7
02/20/2025 16:24	817	3.4	02/20/2025 17:27	822	2.8
02/20/2025 16:25	816	3.3	02/20/2025 17:28	822	2.8
02/20/2025 16:26	815	3.3	02/20/2025 17:29	821	2.7
02/20/2025 16:27	816	3.1	02/20/2025 17:30	821	2.9
02/20/2025 16:28	816	3.3	02/20/2025 17:31	822	3.1
			02/20/2025 17:32	822	3.1
			02/20/2025 17:33	822	3.2
			02/20/2025 17:34	823	3.2
			02/20/2025 17:35	823	3.3
			02/20/2025 17:36	822	3.3
			02/20/2025 17:37	823	3.3
			02/20/2025 17:38	821	3.3
			02/20/2025 17:39	822	3.3
			02/20/2025 17:40	823	3.4
			02/20/2025 17:41	823	3.4
			Grand Summaries		
			Avg: 821	Avg: 3.1	
			Count: 136	Count: 136	

Date/Time	BOILER01 LOAD (MW) Value	BOILER01 PM_RAW (mg/scm) Value			
02/20/2025 17:51	822	3.2	02/20/2025 18:54	821	2.9
02/20/2025 17:52	822	3.2	02/20/2025 18:55	820	3.2
02/20/2025 17:53	822	3.2	02/20/2025 18:56	821	3.2
02/20/2025 17:54	823	3	02/20/2025 18:57	821	3.3
02/20/2025 17:55	823	3.3	02/20/2025 18:58	819	3.4
02/20/2025 17:56	822	3.2	02/20/2025 18:59	819	3.4
02/20/2025 17:57	822	3.2	02/20/2025 19:00	819	3.4
02/20/2025 17:58	823	3.2	02/20/2025 19:01	819	3.2
02/20/2025 17:59	822	3	02/20/2025 19:02	817	3.2
02/20/2025 18:00	822	3.1	02/20/2025 19:03	816	3.3
02/20/2025 18:01	822	2.8	02/20/2025 19:04	815	3.4
02/20/2025 18:02	823	2.9	02/20/2025 19:05	816	3.3
02/20/2025 18:03	822	3	02/20/2025 19:06	817	3.2
02/20/2025 18:04	822	2.7	02/20/2025 19:07	817	3.3
02/20/2025 18:05	822	3	02/20/2025 19:08	817	3.4
02/20/2025 18:06	822	2.8	02/20/2025 19:09	819	3.5
02/20/2025 18:07	821	2.9	02/20/2025 19:10	818	3.4
02/20/2025 18:08	821	2.8	02/20/2025 19:11	818	3.4
02/20/2025 18:09	821	2.7	02/20/2025 19:12	819	3.6
02/20/2025 18:10	821	2.9	02/20/2025 19:13	820	3.5
02/20/2025 18:11	821	2.7	02/20/2025 19:14	820	3.6
02/20/2025 18:12	822	2.9	02/20/2025 19:15	821	3.5
02/20/2025 18:13	821	3.1	02/20/2025 19:16	821	3.2
02/20/2025 18:14	821	3.1	02/20/2025 19:17	820	3.3
02/20/2025 18:15	821	3.2	02/20/2025 19:18	820	3.1
02/20/2025 18:16	820	3.3	02/20/2025 19:19	820	3
02/20/2025 18:17	820	3.3	02/20/2025 19:20	821	3.3
02/20/2025 18:18	821	3.3	02/20/2025 19:21	821	3.2
02/20/2025 18:19	821	3.3	02/20/2025 19:22	820	3.3
02/20/2025 18:20	821	3.4	02/20/2025 19:23	821	3.2
02/20/2025 18:21	821	3.4	02/20/2025 19:24	821	3.1
02/20/2025 18:22	821	3.4	02/20/2025 19:25	821	3.2
02/20/2025 18:23	822	3.4	02/20/2025 19:26	820	2.9
02/20/2025 18:24	822	3.5	02/20/2025 19:27	820	3.2
02/20/2025 18:25	823	3.4	02/20/2025 19:28	820	2.9
02/20/2025 18:26	823	3.4	02/20/2025 19:29	820	2.9
02/20/2025 18:27	822	3.4	02/20/2025 19:30	821	3
02/20/2025 18:28	822	3.5	02/20/2025 19:31	818	2.9
02/20/2025 18:29	822	3.5	02/20/2025 19:32	818	3.1
02/20/2025 18:30	822	3.5	02/20/2025 19:33	819	2.8
02/20/2025 18:31	827	3.4	02/20/2025 19:34	819	2.9
02/20/2025 18:32	822	3.4	02/20/2025 19:35	818	2.9
02/20/2025 18:33	817	3.2	02/20/2025 19:36	816	2.8
02/20/2025 18:34	821	3.2	02/20/2025 19:37	815	3.1
02/20/2025 18:35	824	3.3	02/20/2025 19:38	814	3.2
02/20/2025 18:36	826	3	02/20/2025 19:39	816	3.2
02/20/2025 18:37	824	3.2	02/20/2025 19:40	814	3.3
02/20/2025 18:38	822	3.2	02/20/2025 19:41	814	3.2
02/20/2025 18:39	822	3.1	02/20/2025 19:42	814	3.2
02/20/2025 18:40	822	3.3	02/20/2025 19:43	814	3.1
02/20/2025 18:41	822	3	02/20/2025 19:44	815	3.3
02/20/2025 18:42	822	3.1	02/20/2025 19:45	816	3.5
02/20/2025 18:43	823	3	02/20/2025 19:46	816	3.5
02/20/2025 18:44	822	2.8	02/20/2025 19:47	815	3.5
02/20/2025 18:45	822	3.1	02/20/2025 19:48	817	3.4
02/20/2025 18:46	822	2.8	02/20/2025 19:49	819	3.5
02/20/2025 18:47	822	2.9	02/20/2025 19:50	817	3.5
02/20/2025 18:48	821	2.8	02/20/2025 19:51	818	3.4
02/20/2025 18:49	822	2.7	02/20/2025 19:52	818	3.4
02/20/2025 18:50	822	3	02/20/2025 19:53	820	3.5
02/20/2025 18:51	822	2.7	02/20/2025 19:54	820	3.4
02/20/2025 18:52	822	3.1	02/20/2025 19:55	820	3.4
02/20/2025 18:53	823	2.9	02/20/2025 19:56	821	3.4
			02/20/2025 19:57	821	3.4
			02/20/2025 19:58	823	3.3
			02/20/2025 19:59	823	3.3
			02/20/2025 20:00	822	3.2
			02/20/2025 20:01	822	3
			02/20/2025 20:02	822	3.4
			02/20/2025 20:03	822	3.2
			02/20/2025 20:04	822	3.3
			02/20/2025 20:05	822	3.2
			02/20/2025 20:06	822	3.1
			Grand Summaries		
				Avg: 820	Avg: 3.2
				Count: 136	Count: 136

Date/Time	BOILER01 LOAD (MW) Value	BOILER01 PM_RAW (mg/scm) Value	Date/Time	BOILER01 LOAD (MW) Value	BOILER01 PM_RAW (mg/scm) Value
02/21/2025 10:55	500	3.1	02/21/2025 12:24	501	16.1
02/21/2025 10:56	500	2.7	02/21/2025 12:25	501	15.9
02/21/2025 10:57	501	2.9	02/21/2025 12:26	501	15.4
02/21/2025 10:58	502	2.8	02/21/2025 12:27	500	15.5
02/21/2025 10:59	502	7.3	02/21/2025 12:28	500	15
02/21/2025 11:00	500	13.8	02/21/2025 12:29	500	15.7
02/21/2025 11:01	500	12.1	02/21/2025 12:30	499	15.5
02/21/2025 11:02	500	10.7	02/21/2025 12:31	499	15.9
02/21/2025 11:03	500	8.5	02/21/2025 12:32	498	16.2
02/21/2025 11:04	501	6.9	02/21/2025 12:33	499	16.1
02/21/2025 11:05	501	6.1	02/21/2025 12:34	499	15.9
02/21/2025 11:06	501	5	02/21/2025 12:35	499	15.8
02/21/2025 11:07	501	4.8	02/21/2025 12:36	498	15.7
02/21/2025 11:08	500	5.2	02/21/2025 12:37	499	15.8
02/21/2025 11:09	499	5.2	02/21/2025 12:38	500	15.8
02/21/2025 11:10	499	5.1	02/21/2025 12:39	501	15.4
02/21/2025 11:11	499	4.9	02/21/2025 12:40	501	15.9
02/21/2025 11:12	500	4.5	02/21/2025 12:41	501	16
02/21/2025 11:13	501	4	02/21/2025 12:42	501	15.9
02/21/2025 11:14	502	6.3	02/21/2025 12:43	500	15.9
02/21/2025 11:15	501	7.8	02/21/2025 12:44	500	15.8
02/21/2025 11:16	500	7.2	02/21/2025 12:45	500	15.5
02/21/2025 11:17	500	6.2	02/21/2025 12:46	500	15.5
02/21/2025 11:18	501	5.5	02/21/2025 12:47	501	15.5
02/21/2025 11:19	500	5.1	02/21/2025 12:48	500	15.7
02/21/2025 11:20	500	5.1	02/21/2025 12:49	500	17
02/21/2025 11:21	500	6.5	02/21/2025 12:50	500	17.1
02/21/2025 11:22	499	8.4	02/21/2025 12:51	500	16.7
02/21/2025 11:23	499	8.5	02/21/2025 12:52	500	16
02/21/2025 11:24	499	8.3	02/21/2025 12:53	500	15.2
02/21/2025 11:25	500	7.7	02/21/2025 12:54	500	15.4
02/21/2025 11:26	500	11.5	02/21/2025 12:55	500	15.2
02/21/2025 11:27	500	17.8	02/21/2025 12:56	500	15.4
02/21/2025 11:28	501	17.6	02/21/2025 12:57	500	16.1
02/21/2025 11:29	500	19.1	02/21/2025 12:58	500	17.4
02/21/2025 11:30	501	19.4	02/21/2025 12:59	500	18
02/21/2025 11:31	501	18.5	02/21/2025 13:00	500	17.3
02/21/2025 11:32	500	17.8	02/21/2025 13:01	500	16.6
02/21/2025 11:33	500	17.3	02/21/2025 13:02	501	15.9
02/21/2025 11:34	501	17.3	02/21/2025 13:03	501	15.2
02/21/2025 11:35	501	16.2	02/21/2025 13:04	501	15.7
02/21/2025 11:36	499	16.7	02/21/2025 13:05	501	15
02/21/2025 11:37	499	18.1	02/21/2025 13:06	500	16.2
02/21/2025 11:38	499	17.2	02/21/2025 13:07	499	17.5
02/21/2025 11:39	499	16.7	02/21/2025 13:08	500	17.3
02/21/2025 11:40	499	17	02/21/2025 13:09	500	17.5
02/21/2025 11:41	499	16.3	02/21/2025 13:10	501	16.8
02/21/2025 11:42	499	18	02/21/2025 13:11	500	16.7
02/21/2025 11:43	499	18.3	02/21/2025 13:12	499	16.6
02/21/2025 11:44	500	18.2	02/21/2025 13:13	499	16.7
02/21/2025 11:45	499	17.5	02/21/2025 13:14	501	16.9
02/21/2025 11:46	499	16.9	02/21/2025 13:15	502	16.9
02/21/2025 11:47	499	17.1	02/21/2025 13:16	501	16.8
02/21/2025 11:48	498	16.2	02/21/2025 13:17	501	16.6
02/21/2025 11:49	498	16.6	02/21/2025 13:18	501	16.8
02/21/2025 11:50	498	16.5	02/21/2025 13:19	501	16.5
02/21/2025 11:51	498	16.1	02/21/2025 13:20	501	16.4
02/21/2025 11:52	498	16.1	02/21/2025 13:21	501	16.4
02/21/2025 11:53	500	15.6	02/21/2025 13:22	502	16.4
			02/21/2025 13:23	501	16
Grand Summaries			Grand Summaries		
	Avg: 500	Avg: 11.5		Avg: 500	Avg: 16.2
	Count: 59	Count: 59		Count: 60	Count: 60

Date/Time	BOILER01 LOAD (MW) Value	BOILER01 PM_RAW (mg/scm) Value	Date/Time	BOILER01 LOAD (MW) Value	BOILER01 PM_RAW (mg/scm) Value
02/21/2025 14:04	508	17	02/21/2025 15:30	817	3.2
02/21/2025 14:05	514	16.9	02/21/2025 15:31	818	3.2
02/21/2025 14:06	520	16.6	02/21/2025 15:32	818	3.2
02/21/2025 14:07	523	16.6	02/21/2025 15:33	817	3.2
02/21/2025 14:08	526	16.8	02/21/2025 15:34	817	3.3
02/21/2025 14:09	528	16.8	02/21/2025 15:35	818	3.3
02/21/2025 14:10	530	17.8	02/21/2025 15:36	818	3.3
02/21/2025 14:11	532	18.1	02/21/2025 15:37	819	3.3
02/21/2025 14:12	535	18	02/21/2025 15:38	819	3.3
02/21/2025 14:13	540	17.8	02/21/2025 15:39	818	3.3
02/21/2025 14:14	546	17.9	02/21/2025 15:40	818	3.4
02/21/2025 14:15	551	17.5	02/21/2025 15:41	818	3.1
02/21/2025 14:16	556	17.2	02/21/2025 15:42	816	2.8
02/21/2025 14:17	564	17.2	02/21/2025 15:43	815	3
02/21/2025 14:18	568	17.3	02/21/2025 15:44	817	2.9
02/21/2025 14:19	573	16.9	02/21/2025 15:45	814	2.8
02/21/2025 14:20	579	17.1	02/21/2025 15:46	815	3.1
02/21/2025 14:21	586	17.6	02/21/2025 15:47	814	2.9
02/21/2025 14:22	591	17.5	02/21/2025 15:48	815	3.1
02/21/2025 14:23	594	17.6	02/21/2025 15:49	812	3
02/21/2025 14:24	600	17.8	02/21/2025 15:50	814	3
02/21/2025 14:25	606	17.3	02/21/2025 15:51	817	3
02/21/2025 14:26	613	17.5	02/21/2025 15:52	813	2.7
02/21/2025 14:27	620	17.3	02/21/2025 15:53	813	3
02/21/2025 14:28	624	17	02/21/2025 15:54	818	2.8
02/21/2025 14:29	627	16.9	02/21/2025 15:55	817	2.7
02/21/2025 14:30	632	16.5	02/21/2025 15:56	817	2.8
02/21/2025 14:31	641	16.9	02/21/2025 15:57	817	2.7
02/21/2025 14:32	647	16.3	02/21/2025 15:58	818	2.9
02/21/2025 14:33	652	17.7	02/21/2025 15:59	818	2.7
02/21/2025 14:34	653	18.5	02/21/2025 16:00	816	2.8
02/21/2025 14:35	657	17.7	02/21/2025 16:01	816	2.8
02/21/2025 14:36	660	18.5	02/21/2025 16:02	817	2.7
02/21/2025 14:37	664	18.1	02/21/2025 16:03	819	2.9
02/21/2025 14:38	669	17.7	02/21/2025 16:04	819	3.1
02/21/2025 14:39	676	17.6	02/21/2025 16:05	818	3.2
02/21/2025 14:40	679	17.8	02/21/2025 16:06	818	3.2
02/21/2025 14:41	684	17.7	02/21/2025 16:07	818	3.1
02/21/2025 14:42	694	17.2	02/21/2025 16:08	818	3.2
02/21/2025 14:43	698	17.6	02/21/2025 16:09	818	3.3
02/21/2025 14:44	700	17.5	02/21/2025 16:10	818	3.3
02/21/2025 14:45	705	17.3	02/21/2025 16:11	818	3.4
02/21/2025 14:46	707	18.8	02/21/2025 16:12	817	3.3
02/21/2025 14:47	713	18.8	02/21/2025 16:13	816	3.3
02/21/2025 14:48	716	18.7	02/21/2025 16:14	816	3.2
02/21/2025 14:49	722	18.6	02/21/2025 16:15	816	3.2
02/21/2025 14:50	725	18.5	02/21/2025 16:16	817	3.1
02/21/2025 14:51	733	18.3	02/21/2025 16:17	818	3.2
02/21/2025 14:52	739	19	02/21/2025 16:18	818	3.3
02/21/2025 14:53	743	20.3	02/21/2025 16:19	817	3.4
02/21/2025 14:54	748	20.3	02/21/2025 16:20	817	3.5
02/21/2025 14:55	753	20	02/21/2025 16:21	820	3.4
02/21/2025 14:56	757	19.6	02/21/2025 16:22	821	3.5
02/21/2025 14:57	763	19.6	02/21/2025 16:23	820	3.4
02/21/2025 14:58	768	19.8	02/21/2025 16:24	819	3.2
02/21/2025 14:59	775	19	02/21/2025 16:25	819	3.3
02/21/2025 15:00	779	20.2	02/21/2025 16:26	820	3.2
02/21/2025 15:01	782	20.7	02/21/2025 16:27	819	2.9
02/21/2025 15:02	788	20.2	02/21/2025 16:28	818	3.2
02/21/2025 15:03	795	20.2	02/21/2025 16:29	818	3.1
02/21/2025 15:04	799	19	02/21/2025 16:30	818	3.1
			02/21/2025 16:31	817	3
			02/21/2025 16:32	817	3
			02/21/2025 16:33	816	3.1
			02/21/2025 16:34	817	2.8
			02/21/2025 16:35	819	2.9
			02/21/2025 16:36	819	3
			02/21/2025 16:37	819	2.8
			02/21/2025 16:38	820	3
			02/21/2025 16:39	821	2.8
			02/21/2025 16:40	823	3
Grand Summaries			Grand Summaries		
	Avg: 650	Avg: 18.0		Avg: 817	Avg: 3.1
	Count: 61	Count: 61		Count: 71	Count: 71

Date/Time	BOILER01 LOAD (MW) Value	BOILER01 PM_RAW (mg/scm) Value
02/21/2025 17:22	818	2.8
02/21/2025 17:23	818	3
02/21/2025 17:24	818	2.8
02/21/2025 17:25	818	3
02/21/2025 17:26	818	2.8
02/21/2025 17:27	819	3
02/21/2025 17:28	818	3.2
02/21/2025 17:29	818	3.2
02/21/2025 17:30	817	3.2
02/21/2025 17:31	818	3.3
02/21/2025 17:32	819	3.3
02/21/2025 17:33	818	3.4
02/21/2025 17:34	817	3.4
02/21/2025 17:35	817	3.4
02/21/2025 17:36	818	3.4
02/21/2025 17:37	817	3.4
02/21/2025 17:38	818	3.4
02/21/2025 17:39	817	3.4
02/21/2025 17:40	818	3.4
02/21/2025 17:41	818	3.5
02/21/2025 17:42	819	3.6
02/21/2025 17:43	818	3.6
02/21/2025 17:44	818	3.5
02/21/2025 17:45	817	3.5
02/21/2025 17:46	818	3.5
02/21/2025 17:47	819	3.5
02/21/2025 17:48	818	3.4
02/21/2025 17:49	818	3.2
02/21/2025 17:50	819	3.3
02/21/2025 17:51	821	3.1
02/21/2025 17:52	819	3.2
02/21/2025 17:53	817	3.3
02/21/2025 17:54	816	3
02/21/2025 17:55	817	3.2
02/21/2025 17:56	817	3.1
02/21/2025 17:57	818	3.2
02/21/2025 17:58	818	3.1
02/21/2025 17:59	817	2.7
02/21/2025 18:00	817	2.9
02/21/2025 18:01	817	2.7
02/21/2025 18:02	817	2.9
02/21/2025 18:03	816	2.9
02/21/2025 18:04	817	2.8
02/21/2025 18:05	817	3
02/21/2025 18:06	818	2.7
02/21/2025 18:07	818	2.9
02/21/2025 18:08	818	2.9
02/21/2025 18:09	818	3
02/21/2025 18:10	819	3.1
02/21/2025 18:11	818	3.2
02/21/2025 18:12	818	3.3
02/21/2025 18:13	818	3.3
02/21/2025 18:14	818	3.3
02/21/2025 18:15	818	3.4
02/21/2025 18:16	817	3.4
02/21/2025 18:17	818	3.4
02/21/2025 18:18	817	3.5
Grand Summaries		
	Avg: 818	Avg: 3.2
	Count: 57	Count: 57

CALIBRATION GAS CERTIFICATIONS



CERTIFICATE OF ANALYSIS / EPA PROTOCOL GAS

Customer & Order Information

PRAXAIR PKG LENEXA KS P
9725 ALDEN RD
LENEXA KS 66215-1128

Certificate Issuance Date: 07/02/2021
Praxair Order Number: 71733991
Part Number: NI CD20015E-AS
Customer PO Number: 79732586

Fill Date: 06/16/2021
Lot Number: 700011167F4
Cylinder Style & Outlet: AS CGA 590
Cylinder Pressure and Volume: 2000 psig 140 ft3

Certified Concentration

Expiration Date:	07/01/2029	NIST Traceable
Cylinder Number:	DT0031874	Expanded Uncertainty
20.1 %	Carbon dioxide	± 0.1 %
20.00 %	Oxygen	± 0.01 %
Balance	Nitrogen	

ProSpec EZ Cert



Certification Information:

Certification Date: 07/01/2021

Term: 96 Months

Expiration Date: 07/01/2029

This cylinder was certified according to the 2012 EPA Traceability Protocol, Document #EPA-600/R-12/531, using Procedure G1. Uncertainty above is expressed as absolute expanded uncertainty at a level of confidence of approximately 95% with a coverage factor k = 2. Do Not Use this Standard if Pressure is less than 100 PSIG.

O2 responses have been corrected for CO2 interference.

Analytical Data:

(R=Reference Standard, Z=Zero Gas, C=Gas Candidate)

1. Component: Carbon dioxide

Requested Concentration: 20.0 %
Certified Concentration: 20.1 %
Instrument Used: MKS 2031
Analytical Method: FTIR
Last Multipoint Calibration: 06/07/2021

First Analysis Data:				Date	07/01/2021		
Z:	0	R:	20.1	C:	20.2	Conc:	20.2
R:	20	Z:	0	C:	20.1	Conc:	20.1
Z:	0	C:	20.1	R:	20	Conc:	20.1
UOM: ppm				Mean Test Assay:		20.1	%

Reference Standard: Type / Cylinder #: GMIS / DT0033247

Concentration / Uncertainty: 19.99 % ± 0.04 %
Expiration Date: 01/02/2028

Traceable to: SRM # / Sample # / Cylinder #: PRM / 3222577.01 / FF27613
SRM Concentration (enter with units) / 20.008 % / ± 0.028
SRM Expiration Date: 04/01/2020

Second Analysis Data:				Date			
Z:	0	R:	0	C:	0	Conc:	0
R:	0	Z:	0	C:	0	Conc:	0
Z:	0	C:	0	R:	0	Conc:	0
UOM: ppm				Mean Test Assay: %			

2. Component: Oxygen

Requested Concentration: 20.0 %
Certified Concentration: 20.00 %
Instrument Used: Servomex 575
Analytical Method: Paramagnetic
Last Multipoint Calibration: 06/03/2021

First Analysis Data:				Date	07/01/2021		
Z:	0	R:	22.5	C:	20	Conc:	20
R:	22.5	Z:	0	C:	20	Conc:	20
Z:	0	C:	20	R:	22.5	Conc:	20
UOM: %				Mean Test Assay: 20 %			

Reference Standard: Type / Cylinder #: GMIS / CC257802

Concentration / Uncertainty: 22.5 % ± 0.0 %
Expiration Date: 12/02/2027

Traceable to: SRM # / Sample # / Cylinder #: 2659a / 71-D-04 / CAL015785
SRM Concentration (enter with units) / 20.72 % / ± 0.043 %
SRM Expiration Date: 08/23/2021

Second Analysis Data:				Date			
Z:	0	R:	0	C:	0	Conc:	0
R:	0	Z:	0	C:	0	Conc:	0
Z:	0	C:	0	R:	0	Conc:	0
UOM: %				Mean Test Assay: %			

Analyzed By

Mike Monnette

Certified By

Edward E Zucal

Information contained herein has been prepared at your request by qualified experts within Praxair Distribution, Inc. While we believe that the information is accurate within the limits of the analytical methods employed and is complete to the extent of the specific analyses performed, we make no warranty or representation as to the suitability of the use of the information for any purpose. The information is offered with the understanding that any use of the information is at the sole discretion and risk of the user. In no event shall the liability of Praxair Distribution, Inc., arising out of the use of the information contained herein exceed the fee established for providing such information.



DocNumber: 532768



Linde Gas & Equipment Inc.
6055 Brent Drive
Toledo OH 43611
Tel: +1 (419) 729-7732
Fax: +1 (419) 729-2411
PGVP ID: F12023

CERTIFICATE OF ANALYSIS / EPA PROTOCOL GAS

Customer & Order Information

LGEPKG LENEXA KS P
9725 ALDEN RD
LENEXA KS 66215-1128

Certificate Issuance Date: 01/26/2023

Linde Order Number: 72317322

Part Number: NI CD10010E-AS

Customer PO Number: 80324183

Fill Date: 01/10/2023

Lot Number: 700013010F7

Cylinder Style & Outlet: AS

Cylinder Pressure and Volume: 2000 psig 140 ft³

Certified Concentration

Expiration Date:	01/26/2031	NIST Traceable
Cylinder Number:	CC199274	Expanded Uncertainty
9.61 %	Carbon dioxide	± 0.10 %
9.98 %	Oxygen	± 0.04 %
Balance	Nitrogen	

ProSpec EZ Cert



Certification Information:

Certification Date: 01/26/2023

Term: 96 Months

Expiration Date: 01/26/2031

This cylinder was certified according to the 2012 EPA Traceability Protocol, Document #EPA-600/R-12/531, using Procedure G1. Uncertainty above is expressed as absolute expanded uncertainty at a level of confidence of approximately 95% with a coverage factor k = 2. Do Not Use this Standard if Pressure is less than 100 PSIG.

O2 responses have been corrected for CO2 interference.

Analytical Data:

(R=Reference Standard, Z=Zero Gas, C=Gas Candidate)

1. Component: Carbon dioxide
Requested Concentration: 10.0 %
Certified Concentration: 9.61 %
Instrument Used: Horiba VA-5000
Analytical Method: NDIR
Last Multipoint Calibration: 12/28/2022

Reference Standard: Type / Cylinder #: GMIS / DT0032770
Concentration / Uncertainty: 20.0 % ± 0.0 %
Expiration Date: 01/02/2028
Traceable to: SRM # / Sample # / Cylinder #: PRM# FF276163 / PRM# FF276163 /
SRM Concentration / Uncertainty: 20.006% / ± 0.028%
SRM Expiration Date: 04/01/2020

First Analysis Data:				Date	01/26/2023
Z:	0	R:	20	C:	9.6
R:	20	Z:	0	C:	9.61
Z:	0	C:	9.61	R:	20.01
UOM:	%	Mean Test Assay:		9.61	%

Second Analysis Data:				Date	
Z:	0	R:	0	C:	0
R:	0	Z:	0	C:	0
Z:	0	C:	0	R:	0
UOM:	%	Mean Test Assay:			%

2. Component: Oxygen
Requested Concentration: 10.0 %
Certified Concentration: 9.98 %
Instrument Used: Servomex 575
Analytical Method: Paramagnetic
Last Multipoint Calibration: 01/06/2023

Reference Standard: Type / Cylinder #: SRM / DT0032356
Concentration / Uncertainty: 22.52 % ± 0.01 %
Expiration Date: 12/02/2027
Traceable to: SRM # / Sample # / Cylinder #: 2659a / 71-D-04 / CAL015785
SRM Concentration / Uncertainty: 20.720% / ± 0.043%
SRM Expiration Date: 06/23/2021

First Analysis Data:				Date	01/26/2023
Z:	0	R:	22.52	C:	9.98
R:	22.51	Z:	0	C:	9.98
Z:	0	C:	9.99	R:	22.52
UOM:	%	Mean Test Assay:		9.98	%

Second Analysis Data:				Date	
Z:	0	R:	0	C:	0
R:	0	Z:	0	C:	0
Z:	0	C:	0	R:	0
UOM:	%	Mean Test Assay:			%

Analyzed By

Lloyd Knapp

Certified By

Edward J. Zucal

Information contained herein has been prepared at your request by qualified experts within Linde Gas & Equipment Inc. While we believe that the information is accurate within the limits of the analytical methods employed and is complete to the extent of the specific analyses performed, we make no warranty or representation as to the suitability of the use of the information for any purpose.

Issue Date: 6/27/2019



To: Westar Energy

Attn: Alan Eason

CERTIFICATE OF CONFORMANCE

Business Confidential

Dear Customer:

This is to advise that UHP Grade Nitrogen, (NI 5.0) supplied to Westar Energy, Topeka, KS by Praxair Distribution, Lenexa, KS meets or exceeds the following minimum purity and maximum impurity specification content:

Cylinder Serial Numbers: SA9069, CC246055, T103969, CC45648

Nitrogen: 99.999% minimum purity

O2: 5 ppm maximum

THC: 0.5 ppm maximum

H2O: 3 ppm maximum

Sincerely,

Todd Hallier
Praxair Distribution Inc.
1608 Holmes
Kansas City MO 64108

EQUIPMENT CALIBRATION RECORDS

Dry Gas Meter Calibration

Current Cal:

$Y_d = 0.986$
$\Delta H_{@} = 1.949$

Previous Cal:

$Y_d = 0.983$
$\Delta H_{@} = 1.930$

Pass: X

Fail:

Console ID:	1806015		
Date:	01/13/25		
Orifice ID:	UZ-48	UZ-55	UZ-63
Orifice K':	0.3413	0.4400	0.5890
	Run 1	Run 2	Run 3
Initial Reading, (ft ³):	191.700	199.800	207.200
Final Reading, (ft ³):	198.832	206.121	214.903
Difference, (ft ³) [V _w]:	7.132	6.321	7.703
Initial Meter Temp., (°F):	60	61	61
Final Meter Temp., (°F):	61	61	62
Average Meter Temp., (°F):	60.5	61.0	61.5
Test Time (min.) [q]:	16.00	11.00	10.00
Orifice Manometer Reading, (in H ₂ O):	0.66	1.15	1.95
Barometric Pressure, (in HG):	29.50	29.50	29.50
Ambient Temperature, (°F):	63	62	62
Pump Vacuum, (in Hg):	23.5	22.5	21
Standard Volume of the Meter, V _{m(std)} =	7.142	6.332	7.724
Standard Volume of Critical Orifice, V _{cr(std)} =	7.044	6.249	7.605
DGM Calibration Factor, (Y) =	0.986	0.987	0.985
$\Delta H_{@}$ =	1.922	2.014	1.912
Percent Difference =	-0.032	-0.105	0.137
Average Y =	0.986	1.000 ± .03 acceptability criteria	
Reference Y _d =	1.000		
Average $\Delta H_{@}$ =	1.949		
STD =	0.00122		
Percent Difference (Previous Cal)	0.30142	± 5% acceptability criteria	

Operator: Brandon Jones

Date: 1/13/2025

Calibration Pyrometer Check		
	Cal Temp °F	Pyrometer Temp °F
Ambient temp	0	3
	59	60.8
	200	201
	250	252
	300	302
	350	351
	400	400
	500	499
	Max Deviation (%)	
		0.7

V&A Instruments Model VA710 Calibrator-Thermometer SN: VA181203023

ET&A ACCREDITATION AND QSTI CERTIFICATIONS



Date: April 1, 2019

To: Environmental Group, Evergy

From: Dan Wilkus, Senior Manager of Environmental Services

Subject: Certification of Compliance with ASTM D 7036-04

As of April 1, 2019, Evergy Testing & Analytics (ET&A) group has self-accredited as an Air Emissions Testing Body (AETB). This letter is to certify that our testing methods and implemented processes will meet the requirements of Federal, State, and local regulations. As required by ASTM D 7036-04 (Standard Practice for Competence of Air Emissions Testing Bodies) and 40 CFR 75 (Continuous Emissions Monitoring) we have developed and emissions testing procedure and will abide by all requirements noted therein.

A handwritten signature in blue ink, appearing to read 'Dan Wilkus', written over a horizontal line.

Dan Wilkus
Senior Manager of Environmental Services

A handwritten signature in blue ink, appearing to read 'Jeremy Duis', written over a horizontal line.

Jeremy Duis
Manager CEMs and Reporting

SOURCE EVALUATION SOCIETY



Qualified Source Testing Individual

LET IT BE KNOWN THAT

BRANDON M. JONES

HAS SUCCESSFULLY PASSED A COMPREHENSIVE EXAMINATION AND SATISFIED
EXPERIENCE REQUIREMENTS IN ACCORDANCE WITH THE GUIDELINES
ISSUED BY THE SES QUALIFIED SOURCE TEST INDIVIDUAL REVIEW BOARD FOR

PART 75 CEMS RATA TESTING

ISSUED THIS 9TH DAY OF APRIL 2024 AND EFFECTIVE UNTIL APRIL 8TH, 2029



CERTIFICATE
NO. 2024-1152

J. Wade Bice
J. Wade Bice, QSTI/QSTO Review Board

Karen D. Kajlya-Mills
Karen D. Kajlya-Mills, QSTI/QSTO Review Board

Bruce Randall
Bruce Randall QSTI/QSTO Review Board

Peter R. Westlin
Peter R. Westlin, QSTI/QSTO Review Board

Peter S. Pakalnis
Peter S. Pakalnis, QSTI/QSTO Review Board

Tina Sanderson
Tina Sanderson, QSTI/QSTO Review Board

PARTICULATE MATTER LAB REPORT



**PARTICULATE MATTER
LAB REPORT**

LA CYGNE GENERATING STATION

ET&A PROJECT 25-012

DATE: 4/20/2025

I certify that to the best of my knowledge all analytical data presented in this report:

- Have been checked for completeness
- Are accurate, error-free, and legible
- Have been conducted in accordance with approved protocol, and that all deviations and analytical problems are summarized in the appropriate narrative

A handwritten signature in purple ink, appearing to read "Paul Jones", is written above a horizontal line.

4/20/2024

Signature / Date

ANALYTICAL NARRATIVE SUMMARY

Chain of Custody

Relinquished By: _____ ON SITE

Date and Time: _____

Received By: _____ ON SITE

Date and Time: _____

Analysis

- The samples were analyzed for particulate matter and condensable particulate matter using the analytical procedures in USEPA Method 5, Determination of Particulate Matter Emissions from Stationary Sources and UPESA Method 202, Determination of Condensable Particulate Matter from Stationary Sources. All samples were analyzed following the procedures in section 11.0, Analytical Procedure.

Calibration

- The analytical balance was calibrated using certified class "S" weights prior to analysis of any samples. For each calibration point, the balance was required to read within 0.5 mg of the actual certified mass of the class "S" weight. The calibration range bracketed the lower and upper mass of the samples. A copy of the calibration log is included in this report.

The analytical environment met the following conditions:

- Temperature: 58-78 °F
- Humidity: < 55%

QC Notes

- An Acetone Blank was prepared and analyzed with the samples. The blank value is included in the Beaker Tare Log of this report.

No analytical discrepancies occurred.

All Samples were processed on a A&D HR-100AZ Analytical Balance with a weighing capacity of 102 grams and readability to 0.0001 grams. The balance is certified annually.

SUMMARY OF RESULTS



EPA Method 5 : Determination of Particulate Emissions from Stationary Sources

Field Data

	Run #1	Run #2	Run #3	Run #4	Run #5
Date:	2/19/2025	2/19/2025	2/19/2025	2/19/2025	2/19/2025
Start time:	10:34	12:00	13:12	14:48	16:00
Stop time:	11:37	13:03	14:15	15:51	17:03

Method 5 Laboratory Data

Beaker Tare Weight (g):	51.7641	51.7811	53.2737	51.9409	53.5973
Beaker Final Weight (g):	51.7688	51.7858	53.2799	51.9464	53.6012
Filterable PM mass (probe) (g):	0.0047	0.0047	0.0062	0.0055	0.0039
Filter Tare Weight (g):	0.3528	0.3522	0.3547	0.3536	0.3579
Filter Final Weight (g):	0.3536	0.3527	0.3549	0.3539	0.3584
Filterable PM mass (filter) (g):	0.0008	0.0005	0.0002	0.0003	0.0005
Acetone Wash (ml):	88	62	60	59	65

Field Data

	Run #6	Run #7	Run #8	Run #9	Run #10
Date:	2/19/2025	2/20/2025	2/20/2025	2/20/2025	2/21/2025
Start time:	17:07	13:00	15:26	17:51	10:55
Stop time:	18:10	15:15	17:41	20:06	11:53

Method 5 Laboratory Data

Beaker Tare Weight (g):	53.3839	53.0772	51.8367	53.2265	52.6687
Beaker Final Weight (g):	53.3897	53.0864	51.8451	53.2357	52.6807
Filterable PM mass (probe) (g):	0.0058	0.0092	0.0084	0.0092	0.0120
Filter Tare Weight (g):	0.3561	0.3552	0.3556	0.3566	0.3555
Filter Final Weight (g):	0.3564	0.3558	0.3565	0.3573	0.3638
Filterable PM mass (filter) (g):	0.0003	0.0006	0.0009	0.0007	0.0083
Acetone Wash (ml):	76	56	68	74	89

Field Data

	Run #11	Run #12	Run #13	Run #14
Date:	2/21/2025	2/21/2025	2/21/2025	2/21/2025
Start time:	12:24	14:04	15:30	17:22
Stop time:	13:23	15:04	16:40	18:18

Method 5 Laboratory Data

Beaker Tare Weight (g):	52.4223	54.1080	51.5006	64.6670
Beaker Final Weight (g):	52.4401	54.1305	51.5042	64.6694
Filterable PM mass (probe) (g):	0.0178	0.0225	0.0036	0.0024
Filter Tare Weight (g):	0.3521	0.3550	0.3545	0.3580
Filter Final Weight (g):	0.3597	0.3644	0.3547	0.3584
Filterable PM mass (filter) (g):	0.0076	0.0094	0.0002	0.0004
Acetone Wash (ml):	76	72	53	78

CALIBRATION LOG

ISO/IEC 17025 Calibration Certificate



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Weight

ID Number 4000022685
Certificate Number 01061080-1
Date of Calibration 25-FEB-2019

SECTION 1: NAME AND ADDRESS OF CUSTOMER

End user
Apex Instruments
204 Technology Park Ln
Fuquay Varina NC 27526

Client
Ohaus Corporation
1900 Polaris Parkway
Box D
Columbus OH 43240

SECTION 2: APPROVED SIGNATORY


Annemarie Love, Metrologist

SECTION 3: PERSON PERFORMING WORK

Robotic Calibration

SECTION 4: CERTIFICATE INFORMATION

Description of Masses: ASTM Weight Set

Accuracy Class	: ASTM E617-13 Class 1	Date Received	: 19-FEB-2019
Order Number	: 0074451-00	Date of Calibration	: 25-FEB-2019
Construction	: One Piece, Two Piece	Date of Issue	: 25-FEB-2019
Material	: Aluminum	Weight Range	: 1mg-2mg
	: Stainless Steel		: 5mg-20g
Serial Number	: 4000022685		

SECTION 5: ENVIRONMENTAL CONDITIONS DURING TEST

Temperature: 21.46 °C Pressure: 757.60 mm Hg Relative Humidity: 48%

SECTION 6: PERTINENT INFORMATION

The Weights listed on this calibration report have been compared to reference mass standards that are traceable to the SI through the National Institute of Standards and Technology under Test No. 684/289871-17.

Reference standards and balances used to perform the calibration are listed in Section 10.

The weights calibrated for this report have been calibrated in accordance with Troemner's calibration process. The calibration performed meets the criteria as described in the current revisions of ASTM E617 and OIML R111.

This calibration also meets specifications as outlined in ISO/IEC 17025, ANSI/NCSL Z540-1-1994, and applicable documents.

This certificate of calibration shall not be reproduced except in full, without the written approval of Troemner, LLC. This certificate of calibration must not be used by the customer to claim product endorsement by NIST, NVLAP or any agency of the U.S. government.

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Weight

ID Number 4000022685
Certificate Number 01061080-1
Date of Calibration 25-FEB-2019

NAME AND ADDRESS OF CUSTOMER

End user

Apex Instruments
204 Technology Park Ln
Fuquay Varina NC 27526

Client

Ohaus Corporation
1900 Polaris Parkway
Box D
Columbus OH 43240

SECTION 7: TRUE MASS (MASS IN VACUUM) CALIBRATION DATA

Nominal Mass Value	Serial Number	True Mass	Density ¹ of Weight	Uncertainty (+ or -)
20 g		19.999980 g	8.0300 g/cm ³	0.015 mg
10 g		9.999985 g	8.0300 g/cm ³	0.010 mg
5 g		5.0000007 g	8.0300 g/cm ³	0.0070 mg
2 g		2.0000008 g	8.0300 g/cm ³	0.0060 mg
2 g *		1.9999994 g	8.0300 g/cm ³	0.0060 mg
1 g		0.9999994 g	8.0300 g/cm ³	0.0060 mg
500 mg		0.5000046 g	7.9500 g/cm ³	0.0025 mg
200 mg		0.2000057 g	7.9500 g/cm ³	0.0025 mg
200 mg *		0.1999996 g	7.9500 g/cm ³	0.0025 mg
100 mg		0.1000005 g	7.9500 g/cm ³	0.0025 mg
50 mg		0.0500061 g	7.9500 g/cm ³	0.0025 mg
20 mg		0.0200033 g	7.9500 g/cm ³	0.0020 mg
20 mg *		0.0199995 g	7.9500 g/cm ³	0.0020 mg
10 mg		0.0100039 g	7.9500 g/cm ³	0.0020 mg
5 mg		0.0050036 g	7.9500 g/cm ³	0.0020 mg
2 mg		0.0020033 g	2.7000 g/cm ³	0.0020 mg
2 mg *		0.0020025 g	2.7000 g/cm ³	0.0020 mg
1 mg		0.0010037 g	2.7000 g/cm ³	0.0020 mg

¹ Density is assumed unless otherwise stated

* Denotes weight is marked with a dot.

ISO/IEC 17025 Calibration Certificate

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Weight

ID Number 4000022685
Certificate Number 01061080-1
Date of Calibration 25-FEB-2019

NAME AND ADDRESS OF CUSTOMER

End user
Apex Instruments
204 Technology Park Ln
Fuquay Varina NC 27526

Client
Ohaus Corporation
1900 Polaris Parkway
Box D
Columbus OH 43240

SECTION 8: CONVENTIONAL MASS CALIBRATION VALUE VS. REFERENCE DENSITY 8000 kg/m³

Nominal Mass Value	Serial Number	Conventional Mass Value	Uncertainty (+ or -)	Tolerance (+ or -)
20 g		19.999992 g	0.015 mg	0.0740 mg
10 g		9.999991 g	0.010 mg	0.0500 mg
5 g		5.0000035 g	0.0070 mg	0.0340 mg
2 g		2.0000019 g	0.0060 mg	0.0340 mg
2 g *		2.0000005 g	0.0060 mg	0.0340 mg
1 g		0.9999999 g	0.0060 mg	0.0340 mg
500 mg		0.5000041 g	0.0025 mg	0.0100 mg
200 mg		0.2000055 g	0.0025 mg	0.0100 mg
200 mg *		0.1999994 g	0.0025 mg	0.0100 mg
100 mg		0.1000004 g	0.0025 mg	0.0100 mg
50 mg		0.0500060 g	0.0025 mg	0.0100 mg
20 mg		0.0200033 g	0.0020 mg	0.0100 mg
20 mg *		0.0199995 g	0.0020 mg	0.0100 mg
10 mg		0.0100039 g	0.0020 mg	0.0100 mg
5 mg		0.0050036 g	0.0020 mg	0.0100 mg
2 mg		0.0020027 g	0.0020 mg	0.0100 mg
2 mg *		0.0020019 g	0.0020 mg	0.0100 mg
1 mg		0.0010034 g	0.0020 mg	0.0100 mg

* Denotes weight is marked with a dot.

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Weight

ID Number 4000022685
Certificate Number 01061080-1
Date of Calibration 25-FEB-2019

NAME AND ADDRESS OF CUSTOMER

End user

Apex Instruments
204 Technology Park Ln
Fuquay Varina NC 27526

Client

Ohaus Corporation
1900 Polaris Parkway
Box D
Columbus OH 43240

SECTION 9: CONVENTIONAL MASS CALIBRATION DATA VS. REFERENCE DENSITY 8000 kg/m³

Nominal Mass Value	Serial Number	Conventional Mass Correction	Uncertainty (+ or -)	Tolerance (+ or -)
20 g		-0.009 mg	0.015 mg	0.0740 mg
10 g		-0.009 mg	0.010 mg	0.0500 mg
5 g		0.0035 mg	0.0070 mg	0.0340 mg
2 g		0.0019 mg	0.0060 mg	0.0340 mg
2 g *		0.0005 mg	0.0060 mg	0.0340 mg
1 g		-0.0001 mg	0.0060 mg	0.0340 mg
500 mg		0.0041 mg	0.0025 mg	0.0100 mg
200 mg		0.0055 mg	0.0025 mg	0.0100 mg
200 mg *		-0.0006 mg	0.0025 mg	0.0100 mg
100 mg		0.0004 mg	0.0025 mg	0.0100 mg
50 mg		0.0060 mg	0.0025 mg	0.0100 mg
20 mg		0.0033 mg	0.0020 mg	0.0100 mg
20 mg *		-0.0005 mg	0.0020 mg	0.0100 mg
10 mg		0.0039 mg	0.0020 mg	0.0100 mg
5 mg		0.0036 mg	0.0020 mg	0.0100 mg
2 mg		0.0027 mg	0.0020 mg	0.0100 mg
2 mg *		0.0019 mg	0.0020 mg	0.0100 mg
1 mg		0.0034 mg	0.0020 mg	0.0100 mg

* Denotes weight is marked with a dot.

ISO/IEC 17025 Calibration Certificate

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Weight

ID Number 4000022685
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NAME AND ADDRESS OF CUSTOMER

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SECTION 10: CALIBRATION PROCEDURE DATA

Nominal Mass Value	Serial Number	Standard Set No.	Cal Due	Balance Used	Cal Due	Procedure Used
20 g		MS002	08/01/19	A200XXL-132	01/01/20	Multi A-B
10 g		C006	08/01/19	A200XXL-133	01/01/20	Multi A-B
5 g		MS002	08/01/19	A200XXL-132	01/01/20	Multi A-B
2 g		A031	07/01/19	A5XL-131	01/01/20	Multi A-B
2 g *		A031	07/01/19	A5XL-131	01/01/20	Multi A-B
1 g		A031	07/01/19	A5XL-131	01/01/20	Multi A-B
500 mg		A031	07/01/19	A5XL-131	01/01/20	Multi A-B
200 mg		A031	07/01/19	A5XL-131	01/01/20	Multi A-B
200 mg *		A031	07/01/19	A5XL-131	01/01/20	Multi A-B
100 mg		A031	07/01/19	A5XL-131	01/01/20	Multi A-B
50 mg		A031	07/01/19	A5XL-131	01/01/20	Multi A-B
20 mg		A031	07/01/19	A5XL-131	01/01/20	Multi A-B
20 mg *		A031	07/01/19	A5XL-131	01/01/20	Multi A-B
10 mg		A031	07/01/19	A5XL-131	01/01/20	Multi A-B
5 mg		A031	07/01/19	A5XL-131	01/01/20	Multi A-B
2 mg		A031	07/01/19	A5XL-131	01/01/20	Multi A-B
2 mg *		A031	07/01/19	A5XL-131	01/01/20	Multi A-B
1 mg		A031	07/01/19	A5XL-131	01/01/20	Multi A-B

* Denotes weight is marked with a dot.

Balance Calibration

Make:	A&D Company Limited
Model:	HR-100AZ
SN:	6A7707793

Weight (g)	Response (g)	% Difference
0.0100	0.0099	-1.010%
0.0200	0.0197	-1.523%
0.0500	0.0499	-0.200%
0.1000	0.0999	-0.100%
0.2000	0.1996	-0.200%
0.5000	0.4998	-0.040%
1.0000	0.9999	-0.010%
2.0000	1.9999	-0.005%
5.0000	4.9997	-0.006%
10.0000	9.9997	-0.003%
20.0000	19.9992	-0.004%
40.0000	39.9993	-0.002%
* 100.0000	100.0044	0.004%

* This balance is equipped with an internal 100g calibration weight for self calibration before every test. The 100g weight listed is a check weight provided with the balance.

Internally Calibrated: Yes

Date performed: 1/3/2025

Analyst: Brandon Jones

Calibration due: 1/3/2026

Weight Set SN:
4000022685

TARE WEIGHT LOG

Method 5 Filter Tare Weight

Initial Weight Check		Final Weight Check		Drying Method	
Analyst:	Samuel Burt	Analyst:	Samuel Burt	Desiccator	X
Date:	3/20/2024	Date:	3/22/2024	Oven	X
Time:	16:00	Time:	8:00	Date Into Dryer:	3/19/2024
Lab Temp, °F:	68	Lab Temp, °F:	68	Time Into Dryer:	3 hours
% RH:	35	% RH:	36		

Cal Weight (g)	Weight (g)	Cal Weight (g)	Weight (g)	Balance Information	
0.0500 g	0.0499	0.1000 g	0.1002	Make	A&D
0.1000 g	0.1000	1.0000 g	1.0000	Model	HR-100AZ
0.5000 g	0.4999			Serial #	6A7707793
1.0000 g	0.9998			Calibration Weight Set SN:	
3.0000 g	2.9994			4000022685	

Filter ID	Weight (g)	Weight (g)	Average (g)	Result
5B433	0.3527	0.3529	0.3528	Pass
5B434	0.3522	0.3522	0.3522	Pass
5B435	0.3546	0.3548	0.3547	Pass

Completed By (Signature) / Date:  05/02/2024

Method 5 Filter Tare Weight

Initial Weight Check		Final Weight Check		Drying Method	
Analyst:	Samuel Burt	Analyst:	Samuel Burt	Desiccator	X
Date:	11/26/2024	Date:	11/27/2024	Oven	X
Time:	13:19	Time:	11:46	Date Into Dryer:	11/25/2024
Lab Temp, °F:	60	Lab Temp, °F:	61	Time Into Dryer:	11:24
% RH:	45	% RH:	41		

Cal Weight (g)	Weight (g)	Cal Weight (g)	Weight (g)	Balance Information	
0.0500 g	0.0500	0.1000 g	0.1000	Make	A&D
0.1000 g	0.9999	1.0000 g	1.0001	Model	HR-100AZ
0.5000 g	0.4998			Serial #	6A7707793
1.0000 g	1.0002			Calibration Weight Set SN:	
3.0000 g	2.9996			4000022685	

Filter ID	Weight (g)	Weight (g)	Average (g)	Result
5B440	0.3538	0.3533	0.3536	Pass
5B441	0.3578	0.3579	0.3579	Pass
5B442	0.3559	0.3563	0.3561	Pass
5B443	0.3552	0.3552	0.3552	Pass
5B444	0.3554	0.3558	0.3556	Pass
5B445	0.3564	0.3568	0.3566	Pass
5B446	0.3553	0.3556	0.3555	Pass
5B447	0.3521	0.3521	0.3521	Pass
5B448	0.3551	0.3549	0.3550	Pass
5B449	0.3545	0.3544	0.3545	Pass
5B450	0.3578	0.3582	0.3580	Pass
5B451	0.3534	0.3536	0.3535	Pass

Completed By (Signature) / Date:



11/27/2024

Method 5 Beaker Tare Weight

Initial Weight Check		Final Weight Check		Drying Method	
Analyst:	Brandon Jones	Analyst:	Brandon Jones	Desiccator	X
Date:	2/17/2025	Date:	2/19/2025	Oven	
Time:	17:15	Time:	8:30	Date Into Dryer:	2/14/2025
Lab Temp, °F:	64	Lab Temp, °F:	63.2	Time Into Dryer:	20:30
% RH:	32	% RH:	18		

Cal Weight (g)	Weight (g)	Cal Weight (g)	Weight (g)	Balance Information	
10.0000 g	9.9994	40.0000 g	39.9991	Make	A&D
20.0000 g	19.9991	100.0000 g	100.0045	Model	HR-100AZ
30.0000 g	29.9992			Serial #	6A7707793
40.0000 g	39.9989			Calibration Weight Set SN:	
100.0000 g	100.0044			4000022685	

Beaker ID	Weight (g)	Weight (g)	Average (g)	Result
1	51.7638	51.7643	51.7641	Pass
2	51.7811	51.7810	51.7811	Pass
3	53.2737	53.2736	53.2737	Pass
4	51.9408	51.9410	51.9409	Pass
5	53.5975	53.5971	53.5973	Pass
6	53.3838	53.3839	53.3839	Pass
7	53.0772	53.0772	53.0772	Pass
8	54.5106	54.5105	54.5106	Pass
9	53.4210	53.4208	53.4209	Pass
10	51.8368	51.8366	51.8367	Pass
11	54.0815	54.0814	54.0815	Pass
12	54.5821	54.5823	54.5822	Pass
13	53.2264	53.2265	53.2265	Pass
14	52.8931	52.8929	52.8930	Pass
15	52.3769	52.3768	52.3769	Pass
16	52.6687	52.6687	52.6687	Pass
17	52.4223	52.4222	52.4223	Pass
18	54.1082	54.1078	54.1080	Pass
19	51.5005	51.5006	51.5006	Pass
20	64.6669	64.6670	64.6670	Pass
Blank (21)	52.4024	52.4022	52.4023	Pass

Completed By (Signature) / Date:  02/19/2025

ANALYTICAL WEIGHT LOG

Method 5 Filter Analytical Weight

Initial Weight Check		Final Weight Check		Drying Method	
Analyst:	Brandon Jones	Analyst:	Brandon Jones	Desiccator	X
Date:	2/24/2025	Date:	2/24/2025	Oven	
Time:	8:15	Time:	16:00	Date Into Dryer:	2/19-2/21/2025
Lab Temp, °F:	78	Lab Temp, °F:	79	Time Into Dryer:	Multiple
% RH:	24	% RH:	31		

Cal Weight (g)	Weight (g)	Cal Weight (g)	Weight (g)	Balance Information	
0.0500 g	0.0500	0.1000 g	0.1000	Make	A&D
0.1000 g	0.1000	1.0000 g	1.0001	Model	HR-100AZ
0.5000 g	0.5002			Serial #	6A7707793
1.0000 g	1.0000			Calibration Weight Set SN:	
3.0000 g	3.0001			4000022685	

Filter ID	Sample ID	Weight (g)	Weight (g)	Average (g)	Result
5B333	R1	0.3536	0.3536	0.3536	Pass
5B334	R2	0.3526	0.3527	0.3527	Pass
5B335	R3	0.3549	0.3548	0.3549	Pass
5B440	R4	0.3539	0.3538	0.3539	Pass
5B441	R5	0.3584	0.3583	0.3584	Pass
5B442	R6	0.3564	0.3564	0.3564	Pass
5B443	R7	0.3560	0.3556	0.3558	Pass
5B444	R8	0.3563	0.3566	0.3565	Pass
5B445	R9	0.3575	0.3570	0.3573	Pass
5B446	R10	0.3637	0.3638	0.3638	Pass
5B447	R11	0.3597	0.3596	0.3597	Pass
5B448	R12	0.3643	0.3644	0.3644	Pass
5B449	R13	0.3548	0.3546	0.3547	Pass
5B450	R14	0.3583	0.3584	0.3584	Pass
5B451	F blank	0.3533	0.3534	0.3534	Pass

Completed By (Signature) / Date:



2/24/2025

Method 5 Beaker Analytical Weight

Initial Weight Check		Final Weight Check		Drying Method	
Analyst:	Brandon Jones	Analyst:	Brandon Jones	Desiccator	X
Date:	2/24/2025	Date:	2/24/2025	Oven	
Time:	8:15	Time:	16:00	Date Into Dryer:	2/19-2/21/2025
Lab Temp, °F:	78	Lab Temp, °F:	79	Time Into Dryer:	Multiple
% RH:	24	% RH:	31		

Cal Weight (g)	Weight (g)	Cal Weight (g)	Weight (g)	Balance Information	
10.0000 g	10.0000	40.0000 g	40.0004	Make	A&D
20.0000 g	20.0001	100.0000 g	100.0059	Model	HR-100AZ
30.0000 g	30.0002			Serial #	6A7707793
40.0000 g	40.0002			Calibration Weight Set SN:	
100.0000 g	100.0057			4000022685	

Beaker ID	Sample ID	Weight (g)	Weight (g)	Average (g)	Result
1	R1	51.7689	51.7687	51.7688	Pass
2	R2	51.7857	51.7859	51.7858	Pass
3	R3	53.2798	53.2799	53.2799	Pass
4	R4	51.9463	51.9464	51.9464	Pass
5	R5	53.6012	53.6012	53.6012	Pass
6	R6	53.3899	53.3895	53.3897	Pass
7	R7	53.0864	53.0863	53.0864	Pass
8	R7W	54.5110	54.5114	54.5112	Pass
9	R7H	53.4217	53.4218	53.4218	Pass
10	R8	51.8453	51.8449	51.8451	Pass
11	R8W	54.0825	54.0823	54.0824	Pass
12	R8H	54.5829	54.5828	54.5829	Pass
13	R9	53.2357	53.2356	53.2357	Pass
14	R9W	52.8938	52.8938	52.8938	Pass
15	R9H	52.3773	52.3772	52.3773	Pass
16	R10	52.6806	52.6807	52.6807	Pass
17	R11	52.4401	52.4400	52.4401	Pass
18	R12	54.1306	54.1304	54.1305	Pass
19	R13	51.5042	51.5041	51.5042	Pass
20	R14	64.6695	64.6693	64.6694	Pass
21	A Blank	52.4021	52.4022	52.4022	Pass

Completed By (Signature) / Date:



2/24/2025